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**Development and Evaluation of an Industry-Specific Hourly
Techno-Economic Decision-Support Tool for Photovoltaic-Thermal Systems**

Thesis submitted as a requirement for the degree of Bachelor of Engineering (Honours) in
Mechatronic Engineering

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ORIGINALITY STATEMENT

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Abstract

Write a short abstract restating the main objectives of the project, your main findings and their significance. Do not exceed 100 words in the abstract.

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Contents

Make sure to delete checkmark at the end.

I. Introduction	1
A. Background and motivation ✓	1
B. Project problem ✓	1
C. Aim ✓	2
D. Research question ✓	2
E. Project Scope, Contributions and Report Organisation ✓	2
II. Literature review	4
A. Photovoltaic-thermal systems ✓	4
B. Environmental effects on PVT performance ✓	5
C. Hourly weather and solar-resource modelling ✓	6
D. PVT thermal and electrical models ✓	8
E. Mains-water temperature modelling ✓	10
F. Industrial energy-demand modelling ✓	11
G. Techno-economic evaluation ✓	13
H. Existing simulation and web tools ✓	14
I. Research gap ✓	14
III. Methodology	16
A. Overall system architecture ✓	16
B. User inputs and assumptions ✓	16
C. Weather-data acquisition ✓	18
D. Solar geometry and irradiance transposition ✓	19
E. PVT supply model ✓	21
F. Australian mains-water temperature model ✓	24
G. Industry-demand models	26
G.1 Dairy ✓	27
G.2 Brewery	29
G.3 Hotel	29
G.4 Aquatic centre	29
G.5 Commercial laundry	29
H. Hourly energy matching ✓	29
I. Economic and emissions model ✓	29
J. Design explorer and system sizing ✓	31
K. Software implementation ✓	33
L. Validation and verification ✓	33
IV. Results and discussion	35
A. Software and model verification	35
B. Mains-water temperature results ✓	35
C. Industry-demand model results	39

D.	Principal case study 1	39
E.	Principal case study 2	39
F.	Cross-industry comparison	39
G.	Sensitivity and sizing analysis	39
H.	Limitations and uncertainty	39
I.	Engineering significance	39
V.	Conclusions	40
A.	Main findings	40
B.	Contributions	40
C.	Practical value	40
D.	Recommendations and future work	40
	References	40
	Appendices	45

Page budget – working note, remove before submission

Chapter	Recommended pages
I. Introduction	3
II. Literature review	12
III. Methodology	21
IV. Results and discussion	12
V. Conclusions	1
Total	49

List of figures

1	Conceptual relationship between seasonal ambient-air and mains-water temperatures in the Burch–Christensen model. The mains-water cycle is offset, reduced in amplitude and delayed relative to the ambient-air cycle. Source: author, based on references 26 and 27.	11
2	Overall architecture and hourly calculation workflow of the CoolSheet PVT Calculator. External geocoding and PVGIS weather services support a Python FastAPI backend, while the browser frontend performs all engineering and financial calculations.	17
3	Solar geometry and irradiance components used to calculate plane-of-array irradiance on the inclined PVT collector. Direct normal irradiance is projected according to the incidence angle, while diffuse-sky and ground-reflected components are determined using isotropic view factors. The three components are summed to obtain the hourly irradiance supplied to the PVT thermal and electrical models. Source: author.	20
4	Energy balance of the PVT collector control volume. Plane-of-array irradiance is absorbed at the aperture; part is carried away by the heat-transfer fluid between inlet and outlet, part is lost to ambient through the lumped loss coefficients, and the remainder leaves the PV laminate as direct-current electricity. Losses are shown as a single combined term because both thermal formulations lump them into fitted coefficients; the separate wind and long-wave terms of ISO 9806 are implemented but zero in the production configuration. Heat extraction lowers the cell temperature, which is the only coupling between the thermal and electrical branches.	23
5	Construction of the dairy thermal and electrical demand profiles. Annual milk throughput is converted into process-water and electricity requirements, distributed using normalised seasonal and hourly factors, and combined with daily mains-water temperature to generate 8,760-hour demand arrays.	27
6	Hourly matching of PVT supply and facility demand over a representative day. In each timestep the overlap between supply and demand is recorded as directly utilised energy; unmet thermal demand is supplied by the auxiliary heater, unmet electrical demand is imported from the grid, and the excess of each stream is respectively unused or exported. Source: author, schematic.	30
7	Design Explorer calculation sequence. Collector area is adjusted until the selected thermal solar-fraction target is approached.	32
8	Modelled cell temperature and alternating-current output with and without PVT cooling, for the clearest day of the Sydney typical meteorological year. Both panels share the hourly time axis and are taken from a single run of the calculator.	35
9	Fitted BC-Aus daily mains-water profiles against the twelve CER monthly reference values for each of the four solar-water-heater reference locations. Agreement is in-sample: the plotted reference values were used to fit the coefficients.	37
10	Calculation workflow of the BC-Aus mains-water temperature model. The user's coordinates select the nearest Australian reference coefficients, while site-specific PVGIS weather supplies the ambient-temperature statistics used to generate a 365-day mains-water profile.	51

List of tables

1	Principal environmental and operating effects on PVT performance.	5
2	Comparison of selected solar-energy assessment tools.	15
3	Principal user-input groups of the CoolSheet PVT Calculator.	18
4	Fitted BC-Aus coefficients used for runtime reference selection.	25
5	Default process-water coefficients, preheating targets and hourly schedules used in the dairy-demand model.	28
6	Verification and validation evidence applied to the principal calculator components.	34
7	Development of the BC-Aus parameterisation against the CER reference profiles. .	36
8	Agreement between BC-Aus and the CER monthly reference profiles.	36
9	Site, PVT system and mains-override input register.	45
10	Industry-demand input register. Process-selection checkboxes are additionally provided for each industry, all selected by default.	46
11	Economics, emissions, finance and sizing input register.	48
12	Thermal-model coefficients, solver configuration, fixed physical constants and numerical thresholds, at full stored precision. All Model A and Model B coefficients are editable in the interface; the constants and thresholds are not.	49
13	Repository and deployment structure of the CoolSheet PVT Calculator.	51
14	Weather-data record structure retained by the CoolSheet PVT Calculator after validation.	52
15	Worked economic example for the shipped dairy scenario, showing the two evaluation paths of Section III.I side by side. The industry path counts only demand-matched energy but does not deduct operating cost; the supply path deducts operating cost but assumes complete utilisation of both streams. The two are therefore not directly comparable, and the supply-path indicators are upper bounds.	54

Nomenclature

(The nomenclature must respect the following order: latin symbols, Greek symbols, acronyms. Within each category of symbols, follow alphabetic order. An example is provided below)

Latin symbols

Symbol	Definition	Unit
A	PVT collector area	m^2
B_{net}	Annual net financial benefit	AUD yr^{-1}
$a_{0,A}$	Model A zero-loss coefficient	–
$a_{1,A}$	Model A inlet-temperature coefficient	$\text{W m}^{-2} \text{K}^{-1}$
$a_{2,A}$	Model A wind-speed coefficient	s m^{-1}
a_1	Model B linear thermal-loss coefficient	$\text{W m}^{-2} \text{K}^{-1}$
a_2	Model B quadratic thermal-loss coefficient	$\text{W m}^{-2} \text{K}^{-2}$
c_A	Installed capital cost per unit collector area	AUD m^{-2}
c_{buy}	Electricity purchase tariff	AUD kWh^{-1}
c_{export}	Electricity feed-in tariff	AUD kWh^{-1}
c_{gas}	Natural-gas price	AUD MJ^{-1}
c_p	Specific heat capacity of the heat-transfer fluid	$\text{J kg}^{-1} \text{K}^{-1}$
C_0	Initial capital expenditure	AUD
$C_{\text{O\&M}}$	Annual operating and maintenance cost	AUD yr^{-1}
D_m	Number of days in month m	days
e_{dairy}	Dairy electricity intensity	$\text{kWh kL}^{-1} \text{milk}$
E_{AC}	Net alternating-current electrical energy	kWh
E_{DC}	Direct-current electrical energy	kWh
E_{demand}	Facility energy demand	kWh
E_{excess}	Energy exceeding simultaneous demand	kWh
E_{matched}	Energy directly matched to simultaneous demand	kWh
E_{ref}	Reference electrical energy at standard test conditions	kWh
E_{unmet}	Demand not met by the PVT system	kWh
EF_{gas}	Natural-gas emissions factor	$\text{kg CO}_2\text{-e GJ}^{-1}$
EF_{grid}	Grid-electricity emissions factor	$\text{kg CO}_2\text{-e kWh}^{-1}$
f_T	Temperature-dependent electrical correction factor	–
G	Plane-of-array solar irradiance	W m^{-2}
G_b	Beam irradiance on the collector plane	W m^{-2}
G_d	Diffuse irradiance on the collector plane	W m^{-2}
G_r	Ground-reflected irradiance	W m^{-2}
i	Discount rate	–
k_p	Process-water coefficient of process p	$\text{L L}^{-1} \text{product}$
L_{sys}	Combined non-inverter direct-current loss fraction	–

Symbol	Definition	Unit
$\dot{m}$	Heat-transfer-fluid mass flow rate	kg s^{-1}
m_{water}	Mass of water heated	kg
M_{avoided}	Avoided operational greenhouse-gas emissions	$\text{t CO}_2\text{-e yr}^{-1}$
N	Analysis period or system lifetime	years
n	Day of year	–
Q_{th}	Useful thermal energy	kWh
Q_{water}	Water-heating energy demand	kWh
$\dot{Q}_{\text{cool}}$	Thermal power extracted from the PV laminate	W
$\dot{Q}_{\text{th}}$	Useful thermal power	W
r	Solar beam ratio or financial rate, where defined locally	–
$r_{\text{O\&M}}$	Annual operating-cost fraction of capital expenditure	–
s_m	Normalised seasonal factor for month m	–
SF_{el}	Electrical solar fraction	–
SF_{th}	Thermal solar fraction	–
T_{a}	Ambient-air temperature	$^{\circ}\text{C}$
T_{cell}	PV cell or module temperature	$^{\circ}\text{C}$
T_{in}	Collector inlet-fluid temperature	$^{\circ}\text{C}$
T_{m}	Mean collector-fluid temperature	$^{\circ}\text{C}$
T_{mains}	Mains-water temperature	$^{\circ}\text{C}$
T_{NOCT}	Nominal operating cell temperature	$^{\circ}\text{C}$
T_{out}	Collector outlet-fluid temperature	$^{\circ}\text{C}$
T_{target}	Target process water temperature	$^{\circ}\text{C}$
U_{L}	Collector overall heat-loss coefficient	$\text{W m}^{-2} \text{K}^{-1}$
u	Wind speed	m s^{-1}
V_{milk}	Annual raw-milk throughput	L yr^{-1}
$V_{p,h}$	Process-water volume of process p in hour h	L
$w_{p,h}$	Normalised hourly weighting of process p	–

Greek symbols

Symbol	Definition	Unit
α	Solar absorptance	–
β	Collector tilt angle	$^{\circ}$
γ	Collector surface azimuth angle	$^{\circ}$
γ_{p}	PV power-temperature coefficient	K^{-1}
δ	Solar declination angle	$^{\circ}$
η_{AC}	AC delivery or inverter efficiency	–
η_{boiler}	Existing boiler efficiency	–
η_{el}	Electrical conversion efficiency	–
η_{inv}	Inverter efficiency	–
η_{PV}	PV module efficiency at standard test conditions	–
η_{th}	Thermal efficiency	–
η_0	Model B zero-loss thermal efficiency	–
ρ_{g}	Ground albedo	–
ρ_{w}	Density of water	kg L^{-1}

Symbol	Definition	Unit
τ	Solar transmittance	—
ϕ	Site latitude	°
ω	Solar hour angle	°

Acronyms

Acronym	Definition
AC	Alternating current
API	Application programming interface
BC-Aus	Australian Burch–Christensen mains-water temperature model
CAPEX	Capital expenditure
CER	Clean Energy Regulator
CIP	Cleaning in place
CO ₂ -e	Carbon dioxide equivalent
CRF	Capital recovery factor
CSV	Comma-separated values
DC	Direct current
DHI	Diffuse horizontal irradiance
DNI	Direct normal irradiance
GHI	Global horizontal irradiance
HTF	Heat-transfer fluid
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LCC	Life-cycle cost
LCOE	Levelised cost of electricity
LCOH	Levelised cost of heat
NOCT	Nominal operating cell temperature
NPV	Net present value
OPEX	Operating expenditure
POA	Plane of array
PV	Photovoltaic
PVGIS	Photovoltaic Geographical Information System
PVT	Photovoltaic-thermal
SAM	System Advisor Model
SOAC	Sydney Olympic Aquatic Centre
SPP	Simple payback period
STC	Standard test conditions
TMY	Typical Meteorological Year
TRNSYS	Transient System Simulation Tool

I. Introduction

A. Background and motivation

Australian commercial and industrial facilities consume both electricity and heat for water heating, cleaning, washing, refrigeration and process operation. Decarbonising these facilities therefore requires renewable alternatives not only for grid electricity, but also for process heat that is commonly supplied using fossil fuels. Industrial process heat accounts for more than 40% of fossil-fuel consumption within Australian industry, with substantial opportunities identified for renewable technologies in suitable low-temperature applications¹.

Photovoltaic-thermal systems provide a potential method of supplying both forms of energy. A PVT collector combines a photovoltaic module with a thermal absorber or heat exchanger, allowing electricity and useful heat to be generated from the same collector area⁴. The recovered heat may be used for water heating, process preheating, swimming pools or as a heat-pump source. PVT is therefore particularly relevant to facilities with simultaneous electrical and low-temperature thermal demand, especially where available roof area is limited.

Public web calculators have made preliminary conventional photovoltaic assessment increasingly accessible. SunSPOT estimates rooftop solar capacity, installation cost and potential savings for Australian households and businesses, while PVWatts estimates the electrical production of a grid-connected PV system using location and basic system inputs^{2,3}. These platforms demonstrate the value of providing understandable early-stage results without requiring specialist simulation software. However, conventional PV tools do not represent the additional thermal, operational and financial interactions associated with PVT systems.

The CoolSheet PVT Calculator was developed to extend this accessible decision-support approach to combined electrical and thermal generation. Its purpose is to help facility operators, consultants and potential customers determine whether a proposed PVT installation appears sufficiently promising to justify more detailed engineering investigation.

B. Project problem

The central problem addressed by this thesis is not simply predicting how much electricity and heat a PVT collector can generate. It is determining how much of that energy can be used by a particular facility and what technical, financial and environmental value it provides.

Annual generation and consumption totals cannot adequately represent this relationship. A facility may have substantial annual heat demand but limited demand during the hours when solar heat

is available. Similarly, annual PV generation may appear comparable with annual electricity consumption even though production and demand occur at different times. PVT feasibility therefore depends on the hourly alignment of local weather, collector output and facility operation⁵.

This timing is particularly important where thermal storage is unavailable. Heat exceeding simultaneous demand cannot be carried forward and provides no direct fuel saving. Electrical generation must similarly be divided between electricity used on site and surplus electricity exported to the grid, which may have a different economic value. An Australian assessment must also consider seasonal mains-water temperature, industry operating schedules, existing heating efficiency, energy tariffs and emissions factors.

Specialist simulation platforms can model many of these processes but may require detailed input data, commercial software or substantial technical expertise. Conversely, accessible PV calculators generally exclude PVT thermal production, industry-specific demand and combined electrical and thermal financial assessment. A need therefore exists for an intermediate tool that provides sufficient hourly detail for preliminary PVT assessment while remaining accessible through a standard web browser.

C. Aim

To develop and critically evaluate a web-based hourly techno-economic decision-support tool that models PVT electrical and thermal supply, industry-specific energy demand, direct hourly energy utilisation, financial performance and emissions reductions under Australian operating conditions.

D. Research question

How can hourly PVT supply modelling and industry-specific demand modelling be integrated into an accessible web-based tool to estimate technically usable energy and economic feasibility for Australian commercial and industrial facilities?

E. Project Scope, Contributions and Report Organisation

This thesis develops and evaluates an early-stage feasibility tool for liquid-based PVT systems in Australian commercial and industrial facilities. The calculator processes location-specific hourly weather data to estimate electrical and thermal generation and compare those outputs with demand for dairy processing, breweries, hotels, aquatic centres and commercial laundries. It reports direct energy utilisation, electricity export, unmet and excess energy, solar fractions, operating-cost savings, financial performance and avoided emissions.

The model assumes that PVT energy is used during the hour in which it is generated. Thermal storage is not included, so heat exceeding simultaneous demand is recorded as unused. Surplus electricity may be exported, while remaining electrical and thermal demand is supplied by the grid and an auxiliary heating system. The calculator is intended for preliminary feasibility screening rather than detailed hydraulic, structural, electrical or equipment design.

The work formed part of the collaborative CoolSheet project involving Jianxiong Fu and Thai An (Evan) Dang. Jianxiong Fu's earlier work contributed to the initial weather-processing, solar-resource and PVT supply-modelling framework⁶. The calculator evaluated in this thesis subsequently integrated Australian mains-water estimation, five industry-demand models, hourly supply–demand matching, techno-economic and emissions calculations, system-sizing functions and user-facing decision outputs, with technical discussions and validation support from Jianxiong Fu, Thai An Dang and the wider project team.

The remainder of this thesis reviews the relevant literature, presents the calculator methodology and implementation, evaluates the resulting models and case studies, and concludes with the principal findings, limitations and recommendations for future development.

II. Literature review

A. Photovoltaic-thermal systems

Photovoltaic modules convert solar radiation into electricity, whereas solar-thermal collectors transfer absorbed solar energy to a fluid for heating. Photovoltaic-thermal collectors combine these functions within one system, allowing electrical and thermal energy to be generated simultaneously from the same collector area⁴. This combined production can improve the use of limited roof space where both electricity and low-temperature heat are required.

Only part of the solar radiation absorbed by a conventional PV module is converted into electricity. The remainder contributes to module heating or is lost to the surroundings. PV performance is normally referenced to standard test conditions, including an irradiance of $1,000 \text{ W m}^{-2}$ and a cell temperature of 25°C . The reference temperature is not a physical operating threshold, as modules regularly operate above it under outdoor conditions.

For crystalline-silicon modules, increasing cell temperature generally produces a small increase in current but a larger reduction in voltage. Maximum electrical power therefore decreases as cell temperature rises. The magnitude of this response is represented by the module power-temperature coefficient, while IEC 60891 provides standard procedures for correcting measured PV characteristics for changes in irradiance and temperature⁷.

A liquid-based PVT collector transfers part of the absorbed heat to water or another heat-transfer fluid circulating behind the PV module. This can reduce temperature-related electrical losses while producing useful heat. However, the electrical improvement is not fixed; it depends on irradiance, ambient conditions, inlet-fluid temperature, flow rate and the thermal connection between the PV laminate and heat exchanger⁴.

TO DO – Figure II.1 placement: immediately after this paragraph (heat extraction from the PV module). Suggested caption: “Operating principle and principal components of a liquid-based photovoltaic-thermal collector. Source: author, based on references 4, 7 and 8.” Show incident radiation, PV laminate, electrical output, heat exchanger, fluid inlet and outlet, insulation, useful heat and environmental losses. Keep to approximately one-third to half a page. Save the final image file into media/ to insert it here.

Liquid-based collectors are generally more suitable than air-based collectors for hot-water and hydronic applications because liquids can transport useful heat through relatively compact pipework. They may be covered to reduce thermal losses at higher operating temperatures or uncovered to favour low-temperature heat recovery and PV cooling. The appropriate design therefore depends on the required delivery temperature and the relative value of electrical and thermal output.

The system considered in this thesis is based on the CoolSheet retrofit configuration. A lightweight

aluminium heat exchanger is installed behind a framed commercial PV module, allowing thermal energy to be recovered without requiring a purpose-built PVT laminate⁸. The principal environmental and operating conditions affecting this configuration are considered in the following section.

B. Environmental effects on PVT performance

PVT performance is governed by both the surrounding weather and the operating condition of the fluid circuit. The same variable can affect electrical and thermal outputs differently, so isolated statements about increasing or decreasing performance require qualification.

Variable increases	Typical thermal effect	Typical electrical effect	Treatment required in assessment
Solar irradiance	More energy becomes available for heat recovery, but collector temperature may rise	Power generally increases, although elevated cell temperature reduces conversion efficiency	Apply the hourly irradiance incident on the collector plane
Ambient temperature	May reduce heat loss when the collector operates above ambient	Generally increases cell temperature and reduces efficiency	Use hourly ambient temperature
Wind speed	Can increase convective thermal loss, especially for uncovered collectors	Can improve electrical output through module cooling	Include in the module-temperature or heat-loss model where supported
Inlet-fluid temperature	Higher inlet temperature generally reduces thermal efficiency	Can reduce the effectiveness of PV cooling	Link the collector model to the water-supply condition
Fluid flow rate	Improves heat removal up to a system-specific range but reduces temperature rise at higher flow	May reduce PV temperature	Treat as an adjustable operating input and consider pumping implications

Table 1: Principal environmental and operating effects on PVT performance.

Higher irradiance normally increases both electrical production and the heat available for recovery, but insufficient heat removal can raise module temperature. Wind may improve PV cooling while simultaneously increasing collector heat loss. Lower inlet-fluid temperatures and suitable flow rates can improve heat extraction, although higher-temperature heat may be more useful to the receiving process^{9,10,11,12}.

These interactions show why PVT output should be calculated using simultaneous weather and operating conditions rather than independent annual averages. They also indicate that combined

energy yield alone is insufficient: recovered heat must be produced at a usable temperature and during periods of demand.

C. Hourly weather and solar-resource modelling

Solar-energy performance changes throughout both the day and the year. Annual or monthly averages can indicate the total solar resource but cannot preserve the timing of irradiance, ambient temperature and wind speed. This limitation is important for PVT assessment because electrical and thermal production must later be compared with facility demand during the same timestep. An hourly simulation represents a standard year using 8,760 weather records and preserves the temporal variation required for supply–demand matching.

Weather data used in solar modelling may originate from ground measurements, satellite observations or atmospheric reanalysis. Ground stations can provide direct local measurements but are limited to the locations and variables monitored. Satellite products provide broader spatial coverage by estimating surface irradiance from cloud, atmospheric and surface conditions. The Bureau of Meteorology, for example, produces gridded global and direct irradiance products from Himawari satellite observations at ten-minute and hourly intervals¹³. Reanalysis products combine observations with numerical weather models and are particularly useful where complete local measurements are unavailable.

Long-term weather records are commonly condensed into a Typical Meteorological Year for annual energy assessment. A TMY contains one value for each hour of a representative year but does not correspond to one actual calendar year. Representative months are selected from a longer historical period and assembled into a continuous twelve-month dataset. The PVGIS TMY generator applies this approach using the solar-radiation database available for the selected location together with reanalysis meteorological variables¹⁴.

TMY data provide an efficient basis for estimating representative annual performance without requiring a simulation of every year in the historical record. However, a TMY is not a weather forecast and should not be used to predict the conditions of a particular future year. The selected months may originate from different years, and unusual heatwaves, prolonged cloud events or other extreme sequences may not be reproduced. Comparisons between PVGIS TMY data and measured weather records also show that agreement varies between locations and meteorological variables¹⁵. TMY-based results should therefore be interpreted as typical long-term estimates rather than guaranteed annual outcomes.

The principal solar-radiation quantities used in collector modelling are global horizontal irradiance,

direct normal irradiance and diffuse horizontal irradiance. GHI is the total irradiance received by a horizontal surface. DNI represents radiation arriving directly from the solar disc on a surface normal to the beam, while DHI represents radiation scattered by the atmosphere and received from the remainder of the sky¹⁶. Cloud cover and atmospheric conditions alter the relative direct and diffuse components even where total horizontal irradiance is similar.

Weather databases normally provide radiation on a horizontal or predefined reference surface, whereas a PVT collector is installed at a user-defined tilt and azimuth. Solar position must therefore be calculated for each timestep from the site coordinates, date and time. The resulting solar elevation, zenith and azimuth determine the angle between the direct beam and the collector surface.

Accurate time alignment is essential. Solar position depends on the physical time represented by each weather record rather than simply the displayed local clock label. PVGIS hourly outputs are reported using defined time conventions and provide an irradiance time offset to support alignment between each irradiance value and the corresponding sun position^{14,17}. An uncorrected timezone or midpoint error can shift the calculated solar profile and distort both collector output and its alignment with demand.

Horizontal irradiance is then transposed onto the collector plane. Plane-of-array irradiance consists of direct radiation projected onto the inclined surface, diffuse radiation received from the sky and radiation reflected from the ground. Its magnitude therefore depends on solar position, collector tilt, collector azimuth and ground reflectance.

Transposition approaches range from simple isotropic-sky models to anisotropic models that separately represent circumsolar and horizon diffuse radiation. More detailed models may improve performance under some sky conditions but require additional assumptions and input data. The appropriate method therefore depends on the purpose of the assessment, available weather variables and required computational complexity. The pvlib modelling framework provides several established transposition alternatives and illustrates how collector orientation affects annual and seasonal irradiance¹⁸.

Weather-derived estimates remain subject to uncertainty. Gridded datasets represent an area rather than the exact conditions at one roof and may not capture local shading, short-duration cloud behaviour, unusual terrain or site-specific wind exposure. These limitations do not prevent their use in preliminary feasibility assessment, but they must be recognised when interpreting hourly and annual outputs. The specific data-acquisition, time-alignment and transposition methods implemented in the calculator are described in Sections III.C and III.D.

D. PVT thermal and electrical models

PVT modelling requires linked descriptions of thermal and electrical output. The thermal model estimates the solar energy transferred to the circulating fluid, while the electrical model estimates photovoltaic power after accounting for irradiance, module temperature and system losses. The two outputs are coupled because heat extraction influences PV temperature, while fluid and absorber temperatures affect collector heat loss.

Thermal-performance approaches. The useful thermal output of a collector can be represented as absorbed solar energy minus heat losses to the surroundings. One approach is a physical energy balance across the PV laminate, absorber, fluid channels, insulation and external surfaces. Detailed nodal or computational models can represent conduction between layers, thermal capacitance, fluid behaviour and transient heat transfer. These models are valuable for collector design but require geometry, material properties and heat-transfer coefficients that may be unavailable during preliminary assessment.

An alternative is to use coefficients obtained through standardised collector testing. ISO 9806:2025 specifies methods for assessing the durability, reliability, safety and thermal performance of fluid-heating solar collectors¹⁹. Within coefficient-based models, zero-loss efficiency represents performance when the collector operating temperature is close to ambient temperature. First-order and second-order loss coefficients then represent the reduction in efficiency as the mean collector-fluid temperature rises above ambient.

A linear formulation uses a zero-loss efficiency and a first-order heat-loss term. It is computationally efficient and requires relatively few inputs, making it appropriate where only limited collector information is available. Its limitation is that it may not accurately represent the nonlinear increase in heat loss at larger temperature differences.

A quadratic formulation adds a second-order heat-loss coefficient. This generally provides a stronger representation over a wider operating range but requires an additional test parameter. Quasi-dynamic formulations can also include wind dependence, long-wave radiation and collector thermal capacitance. IEA reviews of PVT characterisation note that the appropriate level of model complexity depends on collector type, available test data and the intended application²⁰.

Once thermal efficiency has been estimated, useful thermal power is calculated from the collector area and plane-of-array irradiance. The fluid-temperature rise follows from useful heat, heat-transfer-fluid mass flow rate and specific heat capacity. Greater useful heat increases outlet temperature, whereas a larger mass flow generally produces a smaller temperature rise for the same

thermal power. Where efficiency depends on mean fluid temperature, the outlet temperature is not known initially and an iterative solution may be required.

Electrical-performance approaches. PV electrical behaviour may be represented using detailed current–voltage models or simplified rated-power relationships. Single-diode models use equivalent-circuit parameters to reproduce a module’s full current–voltage curve. They provide detailed electrical information but require module-specific parameters and greater computational effort.

For annual feasibility assessment, a power-temperature relationship provides a simpler alternative. Rated DC power is adjusted according to the incident irradiance and the difference between estimated cell temperature and the reference condition. The PVWatts formulation is an example of this approach, using rated power, effective irradiance, cell temperature and the module power-temperature coefficient²².

Cell temperature must be estimated before the temperature correction can be applied. Empirical methods commonly relate module temperature to plane-of-array irradiance, ambient temperature and wind speed. Nominal operating cell temperature methods provide a relatively simple estimate from a manufacturer-specified operating temperature, while alternatives such as the Faiman, Fuentes, Sandia and PVsyst models include different representations of mounting and convective heat transfer²³.

Most standard PV temperature models describe an uncooled module. In a PVT collector, fluid circulation can reduce module temperature relative to an equivalent PV-only installation. A fully coupled model represents this directly through the collector energy balance. A simplified model may instead estimate an uncooled temperature and apply a cooling adjustment based on recovered heat. This reduces input and computational requirements but introduces uncertainty because the relationship between thermal extraction and PV cell temperature depends on collector construction and thermal contact.

Estimated DC energy must then be adjusted for inverter conversion and other system losses to determine net AC output. These losses may include wiring, mismatch, soiling, availability and conversion losses. For preliminary assessment, combined loss factors and user-defined efficiencies provide a transparent alternative to modelling every electrical component separately.

TO DO – Figure II.3 placement: immediately after this paragraph (electrical system losses). Suggested caption: “Coupling of the thermal and electrical calculation pathways within an hourly PVT model. Source: author, based on references 19–23.” Show common hourly inputs splitting into thermal and electrical branches, with heat extraction connected to PV-temperature adjustment. Keep to approximately half a page. Save the final image file into media/ to insert it here.

The principal modelling trade-off is therefore between physical detail and practical accessibility. Detailed transient thermal models and diode-based electrical models are suitable for collector design and specialist simulation. Coefficient-based thermal models and power-temperature electrical models require fewer inputs and are better suited to rapid hourly feasibility assessment. The selected CoolSheet formulations and their implementation are presented in Section III.E.

E. Mains-water temperature modelling

Mains-water temperature is an important boundary condition in water-heating and solar-thermal analysis. For a fixed quantity of water and target process temperature, the required heating energy increases as inlet temperature decreases. In a PVT system, mains-water temperature can also define the collector inlet condition and therefore affect both predicted thermal production and facility demand.

Mains-water temperature varies seasonally and spatially. Contributing factors include source-water temperature, soil conditions, pipe burial depth, residence time, network flow and local environmental exposure. A study of 5,760 cold-water measurements across 1,255 Melbourne locations reported substantial seasonal and spatial variation and demonstrated that the assumed inlet temperature can materially change calculated water-heating demand²⁴.

Where measured data are unavailable, energy models commonly use a fixed annual value, monthly reference temperatures or an empirical relationship with ambient-air or ground temperature. A constant value is simple but cannot reproduce seasonal variation. Monthly profiles preserve broad seasonality but are generally tied to a limited number of reference climates. Detailed hydraulic and heat-transfer models can provide stronger physical representation but require information about the distribution network that is unlikely to be available during early-stage PVT assessment.

Australian building-energy methods have historically represented water-temperature variation using climatic reference regions. The AccuRate hot-water module describes regions associated with Rockhampton, Alice Springs, Sydney and Melbourne, informed by solar-resource and reticulated-water conditions²⁵. This provides standardised regional treatment but does not directly generate a unique daily temperature profile for every address.

Burch and Christensen developed an empirical alternative in which the annual mains-water temperature is represented as a sinusoidal function of ambient climatic conditions²⁶. The model relates the centre of the mains-water curve to annual mean ambient temperature, reduces its seasonal amplitude and introduces a phase lag. These adjustments represent the damping and delay created by heat transfer through the ground and water-distribution infrastructure.

The model was subsequently incorporated into the United States Building America House Simulation Protocols. In that implementation, the daily estimate uses annual mean ambient temperature, the difference between the warmest and coolest monthly mean temperatures, a mean-temperature offset, an amplitude ratio and a phase lag²⁷. Its main advantage is that it generates a continuous daily profile from weather statistics that are readily available to an energy model.

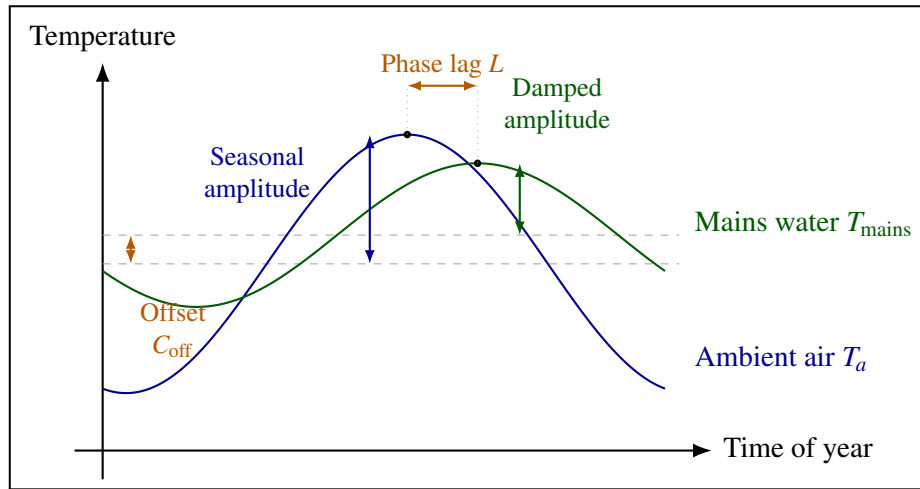


Figure 1: Conceptual relationship between seasonal ambient-air and mains-water temperatures in the Burch–Christensen model. The mains-water cycle is offset, reduced in amplitude and delayed relative to the ambient-air cycle. Source: author, based on references 26 and 27.

Direct application of the original model to Australia presents two issues. First, its calendar convention follows Northern Hemisphere seasonality and would place the Australian annual maximum and minimum approximately six months out of phase. Second, changing the calendar phase alone does not demonstrate that the original offset, amplitude relationship and lag are suitable across Australia’s tropical, arid and temperate climates.

An Australia-wide web calculator therefore requires a method that retains the simplicity of the Burch–Christensen relationship while responding to the climate of the selected site. The method must provide sufficient temporal detail for hourly thermal modelling without requiring information about the local water network. The Australian adaptation developed for this thesis is presented in Section III.F, with its calibration and validation results reported in Section IV.B.

F. Industrial energy-demand modelling

PVT feasibility depends on whether electrical and thermal production occurs when the facility requires energy. Annual consumption cannot represent this timing, so separate hourly electrical and thermal demand profiles are required before direct energy utilisation can be calculated.

Facility demand may be estimated using measured data, industry benchmarks or process-based calculations. Metered electricity, gas and process data provide the strongest representation of an existing facility but may be unavailable during preliminary assessment. Industry benchmarks are easier to obtain but provide limited temporal information and may conceal variation between facilities. Process-based models estimate demand from production activity, water use, target temperatures, equipment power and operating schedules.

Combining process-based calculations with published annual benchmarks used as reasonableness checks provides more temporal detail than applying a single annual intensity, while requiring fewer site-specific data than a complete facility energy audit. This combined approach is adopted in the present work.

TO DO – Figure II.4 placement: immediately after this paragraph (comparing measured, benchmark and process-based methods). Suggested caption: “Development of industry-specific hourly electrical and thermal demand profiles from facility activity, process requirements and operating schedules. Source: author, based on references 1 and 28.” Show facility activity, process quantities, mains-water temperature, target temperatures and schedules leading to separate hourly thermal and electrical profiles. Include small icons for the five industries. Keep to approximately half a page. Save the final image file into media/ to insert it here.

Thermal demand is commonly determined from the quantity of water or material heated, its initial temperature, the required process temperature and the timing of the operation. Mains-water temperature is therefore a direct input because colder incoming water requires more energy to reach the same target temperature. Electrical demand may include refrigeration, pumps, motors, ventilation, lighting and production equipment.

Demand drivers differ between industries. Dairy processing is influenced by milk throughput, cooling and cleaning-in-place requirements; breweries by production volume, hot-water preparation, cleaning and refrigeration; hotels by room occupancy, guest hot water, kitchen and laundry use; aquatic centres by pool size, makeup-water heating, filtration and ventilation; and commercial laundries by linen throughput, washing temperature and cycle timing. Australian process-heat studies and industry guidance confirm that these sectors contain material low-temperature heating opportunities, although individual facility practices vary substantially^{1,28}.

Annual process demand must then be distributed using production days, shifts, occupancy, operating hours, process start times and seasonal factors. The resulting hourly profiles are engineering representations rather than predictions of every facility within a sector. Adjustable user inputs, measured-data calibration and comparison with annual benchmarks are therefore important for managing uncertainty. The detailed equations and schedules for the five implemented industries are presented in Section III.G.

G. Techno-economic evaluation

Techno-economic evaluation converts predicted energy performance into indicators that support investment decisions. For PVT systems, this requires valuing two energy streams: electricity and useful heat. The financial result depends not only on total generation but on the quantity that directly offsets energy purchases.

Electrical savings arise when PVT electricity displaces grid consumption. Electricity generated above simultaneous demand may be exported, but its value may differ from the retail purchase price. A realistic assessment must therefore distinguish electricity used on site from electricity exported and apply the corresponding tariff assumptions.

Thermal savings depend on heat that directly displaces an existing heating source. Where a gas boiler supplies the facility, avoided gas consumption must account for boiler efficiency because more fuel energy is consumed than useful heat delivered. Heat generated when no corresponding demand exists has no direct fuel-saving value when thermal storage is excluded. Hourly matching is consequently necessary to avoid valuing all annual thermal generation as useful.

Capital expenditure includes the collectors, installation, hydraulic equipment, controls and associated electrical components. Operating and maintenance costs, equipment replacement, degradation, project life and residual value may also influence lifetime performance. Australian renewable process-heat studies identify capital cost, energy prices, operating conditions and financial return as important barriers to deployment even where the technical opportunity is favourable¹.

TO DO – Figure II.6 placement (Section II.F claims Figures II.4 and II.5): immediately after this paragraph (the capital-cost paragraph) and before the discussion of financial indicators. Suggested caption: “Translation of hourly PVT energy utilisation into financial and environmental performance indicators. Source: author, based on references 29–31.” Show matched electricity, exported electricity and matched heat leading to avoided purchases, export revenue, displaced fuel, annual savings, payback, NPV, LCOE/LCOH and avoided CO₂-e. Keep to approximately one-third to half a page. Save the final image file into media/ to insert it here.

Simple payback period expresses the time required for cumulative savings to recover the initial investment. It is readily understood but does not account for the time value of money, savings after the payback year or differences in project life. Net present value addresses these limitations by discounting future cash flows to their present value. A positive NPV indicates that discounted benefits exceed discounted costs under the selected assumptions. Established techno-economic software such as SAM reports multiple financial indicators because no single metric completely describes investment performance²⁹.

Levelised cost metrics express lifetime cost per unit of energy. The levelised cost of electricity

is commonly used for electrical generation, while the levelised cost of heat can be applied to useful thermal production. Their interpretation is more complex for PVT because one installation produces two outputs and shared costs must be allocated between them. Levelised cost should therefore be considered alongside annual savings, payback and NPV rather than used alone.

PVT economics are sensitive to collector cost, system size, irradiance, electricity and gas tariffs, export value, demand timing, boiler efficiency, operating expenditure, discount rate and project life. Recent comparative analysis demonstrates that the preferred solar technology can vary substantially with climate, load and market conditions³⁰. Sensitivity and sizing analysis are therefore required to determine which assumptions most strongly influence the predicted outcome.

Avoided operational emissions can be estimated from the grid electricity and heating fuel displaced by the system. Australian assessments can apply the National Greenhouse Accounts Factors for electricity and fuel consumption³¹. These calculations do not constitute a complete life-cycle assessment unless manufacturing, transport, installation and end-of-life impacts are also included.

H. Existing simulation and web tools

Solar-energy assessment tools range from public web calculators to specialist transient simulation environments. They separate broadly into accessible PV-focused calculators, desktop techno-economic software and detailed specialist modelling platforms, as summarised in Table 2.

Public web tools provide rapid and understandable results but generally focus on conventional PV. SAM offers greater technical and financial detail, although combined PVT assessment requires more configuration than selecting a dedicated integrated workflow²⁹. Specialist environments such as Polysun and TRNSYS can represent PVT collectors, storage, heat pumps, controls and hydraulic interactions, but their modelling flexibility is accompanied by greater input, expertise and software requirements^{21,32,33}.

The comparison reveals a trade-off between accessibility and modelling detail. Existing tools provide either rapid PV-focused screening or detailed specialist system simulation, but do not readily combine browser accessibility, Australian hourly weather, five industry-demand models, Australian mains-water estimation, direct hourly PVT matching and integrated financial, emissions and sizing outputs.

I. Research gap

Established methods are available for modelling PVT collectors, hourly weather, industrial energy demand and financial performance. However, these capabilities are generally distributed across

Tool	Access and intended use	Relevant capability	Limitation for the present research problem
SunSPOT	Public Australian web calculator	Rooftop PV and battery potential, costs and savings	Does not model PVT thermal output or industrial heat demand
PVWatts	Public web calculator	Location-based PV electrical generation using limited inputs	Electrical generation only; no PVT heat or industry-demand matching
SAM	Free desktop techno-economic software	Detailed renewable-energy performance and financial models, including PV and solar water heating	PV and solar-water-heating models are separate rather than an integrated industry-specific PVT workflow
Polysun	Professional dynamic simulation software	PVT, storage, heat pumps, hydraulic systems and economic analysis	Requires detailed system definition and specialist software
TRNSYS	Commercial component-based transient environment	Highly configurable PVT, storage, control and building-energy simulations	High expertise, configuration and data burden for early-stage screening

Table 2: Comparison of selected solar-energy assessment tools.

separate tools. Public calculators are accessible but focus mainly on conventional PV, while specialist platforms can model PVT systems in detail but require greater expertise and configuration.

A further gap exists between estimating annual PVT generation and determining how much energy a facility can use directly. A practical preliminary assessment must connect location-specific hourly electrical and thermal supply with industry operating schedules, seasonal mains-water temperature, electricity export, displaced heating fuel and local economic assumptions.

This thesis addresses that gap through an Australian-focused browser tool integrating hourly PVT supply, five industry-demand models, direct supply–demand matching without thermal storage, techno-economic and emissions assessment, and system-sizing analysis.

III. Methodology

A. Overall system architecture

The CoolSheet PVT Calculator was implemented as a modular, web-based platform for the preliminary technical and economic assessment of photovoltaic-thermal systems under Australian commercial and industrial operating conditions. A representative year is simulated using 8,760 hourly timesteps so that energy utilisation is determined from the timing of supply and demand rather than from annual totals alone.

The software uses a decoupled client–server architecture. A static frontend, implemented using HTML, CSS and JavaScript, provides the user interface and performs the principal engineering calculations within the browser. A separate Python backend, implemented using FastAPI, retrieves and preprocesses external weather data, as described in Section III.C. It performs no engineering calculation; its remaining endpoints provide a data-contract health check and optional emailing of a generated report. Confining the backend to data acquisition allows collector size, facility scale and operating assumptions to be varied without repeating the external weather request. Figure 2 summarises the resulting workflow.

Within the browser, the simulation separates into two parallel branches: a supply branch that converts the retrieved weather data into plane-of-array irradiance and then into hourly thermal and alternating-current electrical output (Sections III.D and III.E), and a demand branch that converts facility-scale inputs, process assumptions and operating schedules into hourly thermal and electrical demand for the selected industry (Section III.G).

The BC-Aus mains-water temperature model, described in Section III.F, provides the shared boundary condition between the two branches.

The completed supply and demand arrays are then matched independently at each hourly timestep, and the hourly results aggregated into the annual energy, economic and emissions indicators described in Sections III.H–III.J.

B. User inputs and assumptions

User-defined parameters fall into the four categories summarised in Table 3, presented in the interface as a site and PVT system panel and an industry demand and economics panel, each with collapsible advanced settings. This structure allows the technical system, facility demand and financial conditions to be varied independently. Default values are provided to permit an initial calculation; however, these defaults represent editable engineering scenarios rather than certified industry benchmarks. Results used for case studies were therefore generated from explicitly stated

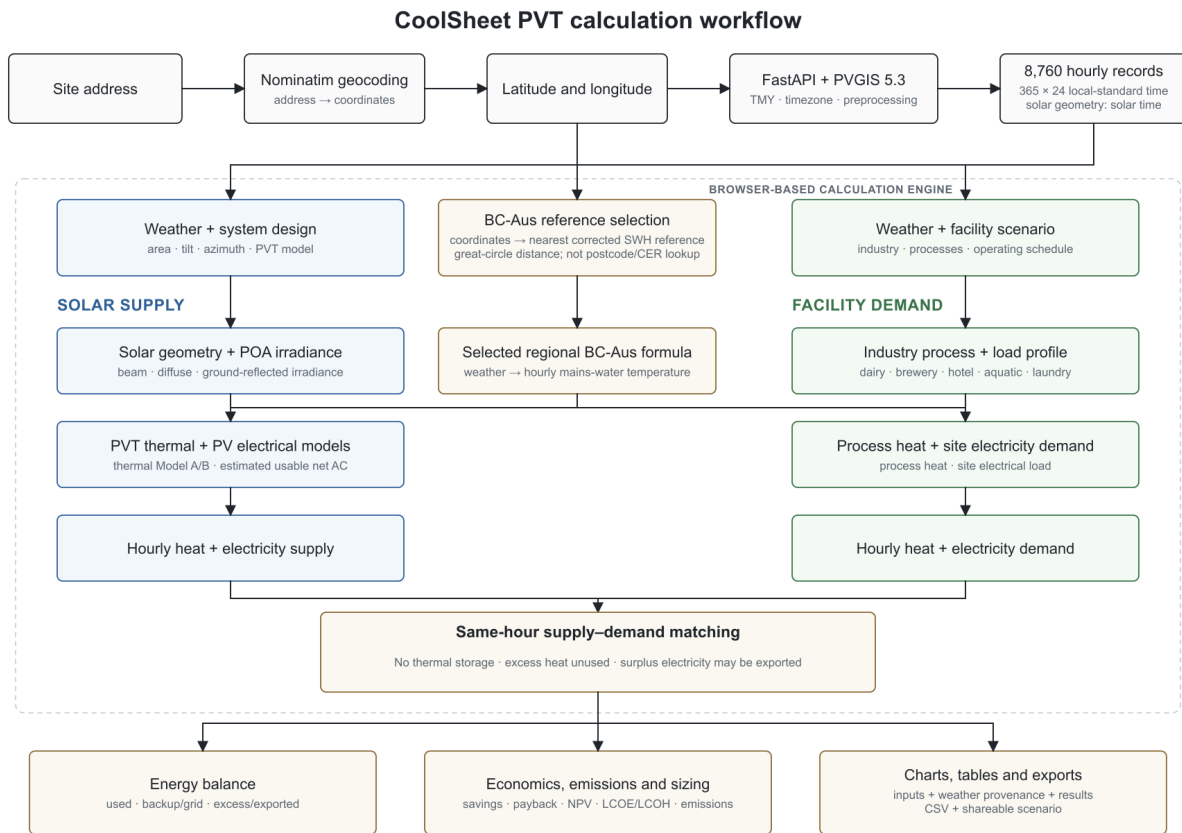


Figure 2: Overall architecture and hourly calculation workflow of the CoolSheet PVT Calculator. External geocoding and PVGIS weather services support a Python FastAPI backend, while the browser frontend performs all engineering and financial calculations.

input sets rather than relying on unspecified interface defaults³⁷. The complete input register, including interface labels, units, shipped defaults and accepted ranges, is provided in Appendix A.

Group	Examples
Site and PVT system	Address, area, tilt, azimuth, flow rate and collector coefficients
Facility demand	Industry, facility scale, process selections and schedule
Economics and emissions	Tariffs, fuel price, boiler efficiency, CAPEX, OPEX and emissions factors
Optional overrides	Monthly mains temperatures and editable model parameters

Table 3: Principal user-input groups of the CoolSheet PVT Calculator.

Several system-level boundaries were fixed to maintain an appropriate level of complexity for preliminary feasibility assessment. The model uses a fixed-tilt homogeneous array and hourly TMY weather³⁵, with site-specific shading, tracking and detailed hydraulic design excluded and wiring, mismatch, soiling and availability effects combined within a single user-defined system-loss factor. Thermal and electrical supply are matched directly to simultaneous demand, with no thermal or battery storage: excess heat is recorded as unused, surplus electricity may be exported, and unmet thermal and electrical demand are assigned to the existing auxiliary heating system and the grid respectively. These boundaries prevent energy generated in one hour from being credited against demand in another, and define the calculator as an early-stage screening model rather than a detailed system-design simulation.

C. Weather-data acquisition

Location-specific weather data were obtained automatically after the user entered a site address. The address was submitted to the OpenStreetMap Nominatim search service and converted into latitude and longitude coordinates, which were retained as the site location for weather retrieval, solar geometry and BC-Aus reference selection³⁴.

The coordinates were transmitted to the calculator’s Python FastAPI backend, which retrieved a Typical Meteorological Year dataset, as defined in Section II.C, from Version 5.3 of the European Commission’s Photovoltaic Geographical Information System^{35,38} using the `get_pvgis_tmy` function provided by the `pvlb` Python library³⁶. The implemented request used a fixed source period from 2005 to 2023, preventing the simulated dataset from changing automatically when additional source years become available³⁷.

For each site, the backend returned 365 days containing 24 hourly records. The principal fields used by the calculator were direct normal irradiance, diffuse horizontal irradiance, global horizon-

tal irradiance, ambient-air temperature and wind speed, together with relative humidity, which is used only by the aquatic-centre outdoor-pool evaporation calculation. Irradiance data provided the inputs to the tilted-surface radiation calculation described in Section III.D, while ambient temperature and wind speed were used by the PVT thermal and electrical models. The ambient-temperature series also supplied the site statistics required by the BC-Aus mains-water temperature model.

The returned timestamps were converted into a synthetic 365-day local-standard-time calendar, with daylight-saving transitions excluded so that every day retained exactly 24 unique hourly records. A separate true-solar-time field was calculated from the UTC timestamp, site longitude and equation of time, and was used only for solar-position calculations; industry operating schedules continued to use local standard clock time. Separating the two time bases prevented daylight-saving transitions, and the difference between clock time and solar time, from shifting either the solar-resource calculation or the facility load profile³⁷. The complete weather-record structure is listed in Appendix B.

D. Solar geometry and irradiance transposition

The weather dataset provides direct normal irradiance and diffuse horizontal irradiance, whereas the PVT models require the total irradiance incident on the inclined collector surface. Solar geometry and irradiance transposition were therefore evaluated independently for each of the 8,760 hourly weather records, using the site latitude, day of year, true solar time, collector tilt, surface azimuth and ground albedo.

Solar declination was estimated using the Cooper correlation³⁹, and the hour angle from true solar time:

$$\delta = 23.45 \sin \left[\frac{360}{365} (n + 284) \right], \quad (1)$$

$$\omega = 15 (h_s - 12), \quad (2)$$

where n is the day of year and h_s is the true solar hour defined in Section III.C, so that $\omega = 0^\circ$ at solar noon.

The solar zenith angle and the incidence angle between the direct beam and the collector normal follow from the site latitude ϕ , declination and hour angle⁴⁰:

$$\cos \theta_z = \sin \delta \sin \phi + \cos \delta \cos \phi \cos \omega, \quad (3)$$

$$\begin{aligned} \cos \theta_i = & \sin \delta \sin \phi \cos \beta - \sin \delta \cos \phi \sin \beta \cos \gamma_s \\ & + \cos \delta \cos \phi \cos \beta \cos \omega + \cos \delta \sin \phi \sin \beta \cos \gamma_s \cos \omega \\ & + \cos \delta \sin \beta \sin \gamma_s \sin \omega, \end{aligned} \quad (4)$$

where β is the collector tilt and γ_s is the surface azimuth. The interface defines azimuth relative to north over the range -180° to $+180^\circ$ (0° north, $+90^\circ$ east, $\pm 180^\circ$ south, -90° west) and subtracts 180° internally to obtain γ_s in the solar-geometry convention, in which 0° is south-facing. The shipped default of 0° therefore specifies a north-facing collector, as appropriate for the Southern Hemisphere^{37,40}.

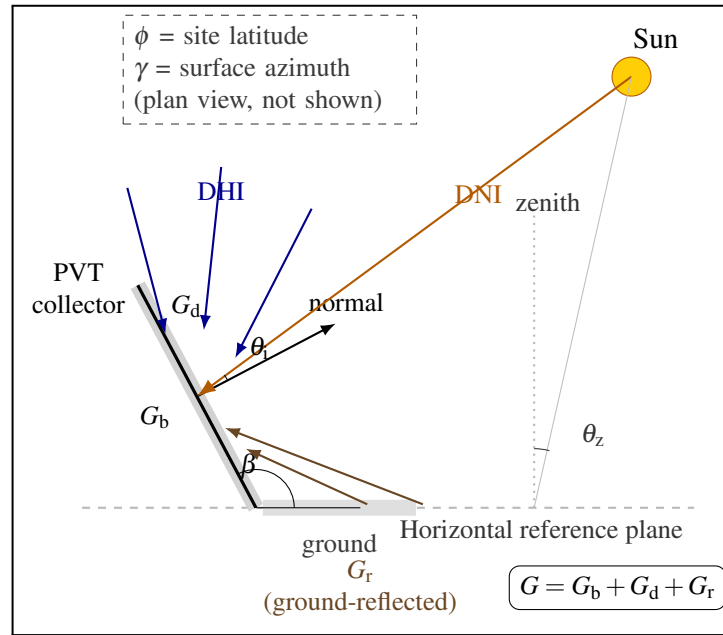


Figure 3: Solar geometry and irradiance components used to calculate plane-of-array irradiance on the inclined PVT collector. Direct normal irradiance is projected according to the incidence angle, while diffuse-sky and ground-reflected components are determined using isotropic view factors. The three components are summed to obtain the hourly irradiance supplied to the PVT thermal and electrical models. Source: author.

Total plane-of-array irradiance was calculated using an isotropic diffuse-sky model⁴¹. The horizontal beam component, reconstructed global horizontal irradiance and the three components incident on the tilted collector were obtained from

$$G_{\text{BHI}} = \text{DNI} \max(0, \cos \theta_z), \quad (5)$$

$$G_{\text{HI}} = G_{\text{BHI}} + \text{DHI}, \quad (6)$$

$$G_{\text{b}} = \text{DNI} \max(0, \cos \theta_i), \quad (7)$$

$$G_{\text{d}} = \text{DHI} \left(\frac{1 + \cos \beta}{2} \right), \quad (8)$$

$$G_{\text{r}} = G_{\text{HI}} \rho_{\text{g}} \left(\frac{1 - \cos \beta}{2} \right), \quad (9)$$

$$G = \max(0, G_{\text{b}} + G_{\text{d}} + G_{\text{r}}), \quad (10)$$

where ρ_{g} is the ground albedo. Direct irradiance was set to zero when the sun was below the horizon or behind the collector plane, and negative irradiance inputs were prevented from propagating through the calculation.

The isotropic model assumes uniform diffuse-sky irradiance and therefore omits circumsolar and horizon-brightening effects. It was retained as the production transposition method because it is transparent, computationally efficient and consistent with the irradiance basis used by the implemented PVT models^{37,41}.

E. PVT supply model

The PVT supply model converted the hourly plane-of-array irradiance G calculated in Section III.D, together with ambient-air temperature T_{a} , wind speed u , mains-water inlet temperature T_{in} , collector area A and heat-transfer-fluid flow rate, into hourly useful thermal energy, outlet-fluid temperature, PVT electrical energy and a conventional PV-only electrical baseline.

Thermal output. Two alternative thermal formulations were implemented. Model A was retained as a project-specific inlet-temperature correlation, whereas Model B applied the ISO 9806 collector-performance formulation^{37,42}. The coefficient sets are model-specific and were not interchanged because Model A uses inlet-fluid temperature while Model B uses mean-fluid temperature. Full-precision coefficients for both models are listed in Appendix A. For Model A, hourly thermal efficiency and useful thermal power were

$$\eta_{th,A} = \text{clamp} \left[a_{0,A} + a_{1,A} \left(\frac{T_{in} - T_a}{G} \right) + a_{2,A} u, 0, 1 \right], \quad (11)$$

$$\dot{Q}_{th} = \eta_{th,A} G A, \quad (12)$$

where $a_{0,A}$ is the zero-loss coefficient, $a_{1,A}$ represents the inlet-temperature loss dependence and $a_{2,A}$ represents the wind-speed dependence. Efficiency and useful power were set to zero when irradiance was negligible.

Model B used the mean fluid temperature $T_m = (T_{in} + T_{out}) / 2$. In the production configuration the optional wind and long-wave coefficients are zero, so the implemented ISO 9806 relation reduces to the quadratic steady-state form⁴²:

$$\dot{Q}_{th} = A \left[\eta_0 G - a_1 (T_m - T_a) - a_2 (T_m - T_a)^2 \right], \quad (13)$$

where η_0 is the zero-loss efficiency and a_1 and a_2 are the linear and quadratic heat-loss coefficients. The remaining ISO 9806 terms are retained within the software so that alternative certified coefficient sets can be applied.

Because T_{out} appears within the mean-fluid temperature, Model B was solved iteratively from the fluid energy balance

$$\dot{m} c_p (T_{out} - T_{in}) - \dot{Q}_{th} (T_{out}) = 0, \quad (14)$$

using Newton–Raphson iteration, with the solver configuration reported in Appendix A³⁷. For Model A, the outlet temperature was calculated directly from the same balance after useful thermal power had been determined. Negative useful thermal power was set to zero for both models.

Electrical output. The reference electrical energy available in each timestep and the uncooled PV cell temperature were calculated as^{40,43}

$$E_{ref} = \frac{\eta_{PV} G A \Delta t}{1000}, \quad (15)$$

$$T_{cell,PV} = T_a + \frac{G}{800} (T_{NOCT} - 20), \quad (16)$$

where η_{PV} is the module efficiency at standard test conditions, $\Delta t = 1$ h and T_{NOCT} is the nominal operating cell temperature. A separate PVT cell temperature represented cooling caused by thermal

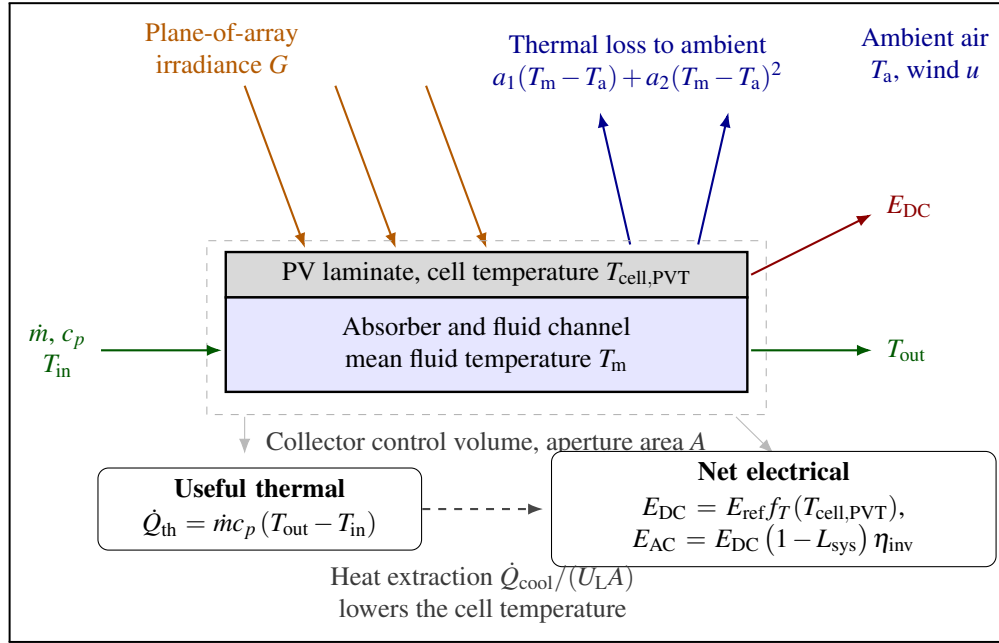


Figure 4: Energy balance of the PVT collector control volume. Plane-of-array irradiance is absorbed at the aperture; part is carried away by the heat-transfer fluid between inlet and outlet, part is lost to ambient through the lumped loss coefficients, and the remainder leaves the PV laminate as direct-current electricity. Losses are shown as a single combined term because both thermal formulations lump them into fitted coefficients; the separate wind and long-wave terms of ISO 9806 are implemented but zero in the production configuration. Heat extraction lowers the cell temperature, which is the only coupling between the thermal and electrical branches.

energy extraction:

$$T_{\text{cell,PVT}} = \text{clamp} \left[T_{\text{cell,PV}} - \frac{\dot{Q}_{\text{cool}}}{U_L A}, T_{\text{in}}, T_{\text{cell,PV}} \right], \quad (17)$$

where $\dot{Q}_{\text{cool}} = \dot{m} c_p (T_{\text{out}} - T_{\text{in}})$ and U_L is the absolute first-order heat-loss coefficient of the active thermal model³⁷.

The temperature-dependent correction and the conversion to net alternating-current energy were

$$f_T = \max [0, 1 + \gamma_P (T_{\text{cell}} - 25)], \quad (18)$$

$$E_{\text{DC,PVT}} = E_{\text{ref}} f_T (T_{\text{cell,PVT}}), \quad E_{\text{DC,PV}} = E_{\text{ref}} f_T (T_{\text{cell,PV}}), \quad (19)$$

$$E_{\text{AC}} = E_{\text{DC}} (1 - L_{\text{sys}}) \eta_{\text{inv}}, \quad (20)$$

where γ_P is the module power-temperature coefficient, L_{sys} is the combined non-inverter direct-current loss fraction and η_{inv} is the inverter efficiency. Net alternating-current PVT energy was

used for the reported electricity generation, hourly demand matching and economic calculations, while the PV-only result was retained as a reference for the estimated electrical effect of thermal cooling.

The cooling relationship is a bounded sensitivity model rather than a detailed cell–absorber heat-transfer model. It was not treated as independently field-validated, and it can be disabled so that PVT and PV-only electrical production are identical. Figure 4 summarises the resulting control volume and the coupling between the two branches.

F. Australian mains-water temperature model

A location-responsive mains-water temperature model was developed because incoming water temperature affects both sides of the PVT assessment. It provides the collector inlet-temperature condition used in the thermal supply model and the initial water temperature used to calculate process-heating demand. A single constant value would therefore introduce the same climatic simplification into both the predicted supply and the facility demand.

Model formulation. The BC-Aus model retained the sinusoidal structure of the original Burch–Christensen relationship. For each day of the representative year, the mains-water temperature is calculated as:

$$T_{\text{mains,F}}(d) = (T_{\text{a,F}} + C_{\text{off}}) + R \left(\frac{\Delta T_{\text{a,F}}}{2} \right) \sin [0.986 (d_m - 15 - L) - 90^\circ] \quad (21)$$

where:

- $T_{\text{mains,F}}(d)$ is the predicted daily mains-water temperature in °F;
- $T_{\text{a,F}}$ is the annual mean ambient-air temperature;
- $\Delta T_{\text{a,F}}$ is the difference between the warmest and coolest monthly mean ambient temperatures;
- C_{off} is the mean-temperature offset;
- R is the seasonal amplitude ratio;
- L is the phase lag in days;
- d_m is the hemisphere-adjusted model day.

The equation is evaluated in the original Fahrenheit domain and the resulting values are converted to degrees Celsius. The original Building America implementation used an offset of +6°F and climate-dependent expressions for amplitude and lag.

Southern Hemisphere adaptation. The original calendar convention follows Northern Hemisphere seasonality. For Australian locations, the model day was shifted by approximately half a year:

$$d_m = ((d + 181) \bmod 365) + 1 \quad (22)$$

where d is the calendar day of year. The shift is applied only to sites at southern latitudes; it sets the broad seasonal orientation, while the fitted phase lag provides the smaller location-specific timing adjustment.

Australian reference fitting. Monthly cold-water reference values were obtained from five legacy Clean Energy Regulator domestic simulation decks associated with Rockhampton, Alice Springs, Sydney, Melbourne and Canberra. These values are Australian simulation references rather than measured utility mains-water records, and this distinction governs how the model’s agreement statistics must be interpreted in Section IV.B. National fits retaining or removing the original fixed offset could not reproduce differences in seasonal amplitude and timing between the reference climates, so each reference profile was instead fitted separately using three effective parameters: mean-temperature offset, amplitude ratio and phase lag. A deterministic three-parameter procedure was adopted because an earlier five-parameter formulation was rank-deficient for a single fixed ambient series and produced model-equivalent but platform-dependent coefficient sets. The complete fitting derivation, data lineage and the excluded Canberra parameters are provided in the appendix.

Reference location	Offset (°F)	Amplitude ratio	Phase lag (days)
Rockhampton	+5.424	0.802	−6.740
Alice Springs	−0.473	1.061	−8.622
Sydney	−0.377	1.034	−0.086
Melbourne	−0.822	0.973	+0.701

Table 4: Fitted BC-Aus coefficients used for runtime reference selection.

The original +6°F offset remained approximately appropriate only for Rockhampton, while the fitted offsets for the remaining references were close to zero (Table 4).

Location-based runtime calculation. When a user enters an address, the geocoded coordinates are compared with the four runtime reference locations and the geographically nearest is selected using the haversine distance, applying its fitted offset, amplitude ratio and phase lag. This is a nearest-reference engineering approximation rather than an official CER postcode-zone determi-

nation. The selected reference and its distance from the site are retained in the result object, but are shown on screen only when the interface’s diagnostic mode is enabled.

The selected coefficients are combined with climate statistics calculated from the PVGIS Typical Meteorological Year dataset for the actual user-selected site, which supplies the annual mean ambient temperature and the warmest-to-coolest monthly temperature range. Consequently, a site assigned to the Melbourne reference does not simply receive Melbourne’s stored monthly profile; the Melbourne-fitted relationship is applied using the actual site’s climate statistics.

Integration with the calculator. The BC-Aus output contains one temperature for each of the 365 days in the representative year. Each daily value is applied to the corresponding hourly records in the 8,760-hour simulation. Within the PVT thermal model, it defines the collector inlet temperature. Within the industry models, the sensible energy required to heat incoming water is calculated using:

$$Q_{\text{water}} = m_{\text{water}} c_p \max(0, T_{\text{target}} - T_{\text{mains}}) \quad (23)$$

The daily profile is used in the dairy, brewery and commercial-laundry water-heating calculations and in aquatic-centre makeup-water heating. Users may replace the automatically generated profile by entering twelve monthly mains-water temperatures where measured or preferred site data are available; months left blank fall back to the modelled daily value.

G. Industry-demand models

The industry-demand models converted annual facility activity into separate hourly thermal and electrical demand profiles. For each industry, annual demand was first estimated from a physical activity variable, such as product throughput, occupancy or water-surface area. Normalised monthly and hourly weighting factors were then applied to distribute this demand across the 8,760-hour representative year.

For water-heating processes, the hourly thermal demand was calculated from the volume of water used, the daily mains-water temperature and the required process temperature:

$$Q_{p,h} = \frac{V_{p,h} \rho_w c_p \max(0, T_{p,\text{target}} - T_{\text{mains},h})}{3600}, \quad (24)$$

where $Q_{p,h}$ is the thermal demand of process p during hour h in kWh, $V_{p,h}$ is the corresponding water volume in litres, $\rho_w = 1 \text{ kg L}^{-1}$ and $c_p = 4.184 \text{ kJ kg}^{-1} \text{ K}^{-1}$. The zero bound prevents negative heating demand when the mains-water temperature exceeds the selected preheating target.

G.1 Dairy

The dairy model represents a dairy-farm or dairy-shed operating scenario rather than a complete milk-processing factory. It estimates electricity use associated with milking, milk cooling, refrigeration, pumping and cleaning, together with low-temperature water preheating for three selectable thermal processes. Pasteurisation, homogenisation, evaporation, drying and other large-scale manufacturing loads are outside the model boundary.

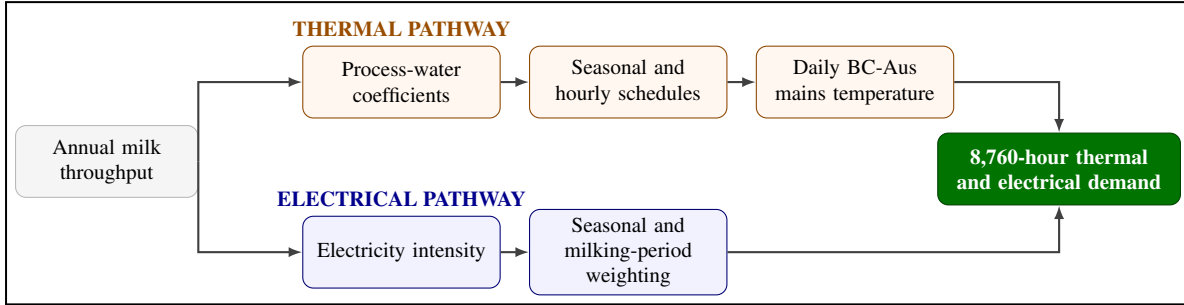


Figure 5: Construction of the dairy thermal and electrical demand profiles. Annual milk throughput is converted into process-water and electricity requirements, distributed using normalised seasonal and hourly factors, and combined with daily mains-water temperature to generate 8,760-hour demand arrays.

Annual raw-milk throughput was used as the principal activity variable. For thermal process p , the annual process-water requirement and its hourly distribution were

$$V_{p,\text{annual}} = V_{\text{milk}} k_p, \quad (25)$$

$$V_{p,h} = \frac{V_{\text{milk}} k_p}{365} s_m w_{p,h}, \quad (26)$$

where V_{milk} is the annual milk throughput in litres, k_p is the process-water coefficient in litres of water per litre of milk, s_m is the normalised seasonal factor for month m and $w_{p,h}$ is the normalised hourly process weighting. The hourly factors for each process sum to unity over a day, while the monthly factors are normalised according to the number of days in each month:

$$\sum_{h=0}^{23} w_{p,h} = 1, \quad \sum_{m=1}^{12} D_m s_m = 365, \quad (27)$$

where D_m is the number of days in month m . These conditions ensure that the annual process-water volume remains equal to $V_{\text{milk}} k_p$ despite the imposed seasonal and hourly variation³⁷. The three implemented thermal processes are summarised in Table 5.

Process	Water-use coefficient	PVT preheating target	Hourly allocation
Fatty-film rinse	0.30 L L ⁻¹ milk	35 °C	50% at 07:00 and 50% at 17:00
CIP preheating	0.57 L L ⁻¹ milk	35 °C	25% at 08:00, 09:00, 17:00 and 18:00
Boiler-feedwater preheating	0.50 L L ⁻¹ milk	35 °C	Uniformly distributed across 24 hours

Table 5: Default process-water coefficients, preheating targets and hourly schedules used in the dairy-demand model.

When all three processes were selected, the default annual heated-water requirement was $k_{\text{total}} = 0.30 + 0.57 + 0.50 = 1.37 \text{ L L}^{-1} \text{ milk}$. The 35 °C target is a single editable input applied to all three processes, and represents the low-temperature preheating contribution that may be supplied directly by the PVT system. It does not represent the final temperature required for full cleaning-in-place or boiler operation, which would normally require additional auxiliary heating.

The raw dairy seasonal factors applied from January to December were 0.85, 0.80, 0.85, 0.90, 0.85, 0.75, 0.75, 0.80, 1.10, 1.30, 1.30 and 1.05, producing lower assumed milk activity during winter and a higher production period during spring. Annual thermal energy was not fixed by the normalisation in Eq. 27 because the daily temperature rise depended on the BC-Aus mains-water profile.

The annual and hourly dairy electrical demand were calculated as

$$E_{\text{el,annual}} = e_{\text{dairy}} \frac{V_{\text{milk}}}{1000}, \quad (28)$$

$$E_{\text{el},h} = \frac{e_{\text{dairy}} V_{\text{milk}}}{365\,000} s_{m w_{\text{el},h}}, \quad (29)$$

where e_{dairy} is the electricity intensity in kWh per kilolitre of milk. The default electricity intensity was 51.7 kWh kL⁻¹, which is within the range reported by Australian dairy-energy audits and close to their reported average^{47,48}. The hourly profile contains a low overnight baseload and elevated demand around the assumed morning and afternoon milking periods, approximately 05:00–07:00 and 15:00–17:00, and represents the combined use of the equipment listed above rather than separate appliance-level simulations. A worked example running from milk throughput through to levelised cost is given in Appendix C.

The dairy model operates continuously across all days of the year. It should therefore be interpreted as a transparent and editable early-stage scenario rather than a certified facility forecast.

The electricity intensity is supported by public Australian benchmark evidence, whereas the exact process-water coefficients and operating schedules remain engineering assumptions based principally on internal project information³⁷. Site-specific assessments should replace these defaults with measured milk throughput, water consumption, operating schedules and interval electricity data where available.

G.2 Brewery

G.3 Hotel

G.4 Aquatic centre

G.5 Commercial laundry

H. Hourly energy matching

Hourly energy matching was used to determine how much of the predicted PVT output could be used directly by the selected facility. Separate 8,760-value arrays were generated for thermal supply, electrical supply, thermal demand and electrical demand, and supply and demand were compared independently within each hourly timestep. For either the thermal or electrical energy stream x , the directly matched energy, unmet demand and excess supply during hour h were

$$E_{\text{matched},x,h} = \min(E_{\text{supply},x,h}, E_{\text{demand},x,h}), \quad (30)$$

$$E_{\text{unmet},x,h} = \max(0, E_{\text{demand},x,h} - E_{\text{supply},x,h}), \quad (31)$$

$$E_{\text{excess},x,h} = \max(0, E_{\text{supply},x,h} - E_{\text{demand},x,h}), \quad (32)$$

which maintain the hourly energy balances $E_{\text{supply},x,h} = E_{\text{matched},x,h} + E_{\text{excess},x,h}$ and $E_{\text{demand},x,h} = E_{\text{matched},x,h} + E_{\text{unmet},x,h}$.

The annual thermal and electrical solar fractions were calculated from the summed hourly values:

$$SF_x = \frac{\sum_{h=1}^{8760} E_{\text{matched},x,h}}{\sum_{h=1}^{8760} E_{\text{demand},x,h}}. \quad (33)$$

Hourly matched, unmet and excess values were also aggregated into monthly totals for presentation and comparison. A separate monthly matching calculation was retained only as an idealised storage upper bound and was not used for the principal no-storage results³⁷.

I. Economic and emissions model

The economic model was evaluated along two separate paths. The industry path valued only the energy classified by the hourly energy balance: electricity consumed directly by the facility at the

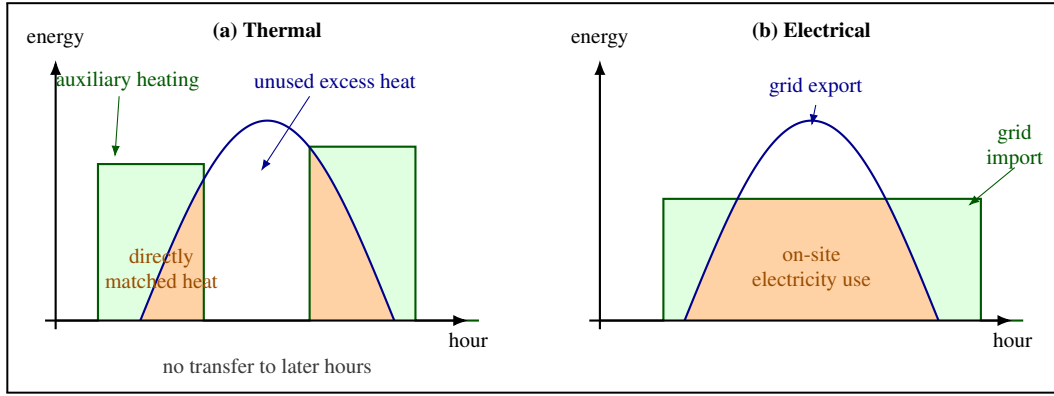


Figure 6: Hourly matching of PVT supply and facility demand over a representative day. In each timestep the overlap between supply and demand is recorded as directly utilised energy; unmet thermal demand is supplied by the auxiliary heater, unmet electrical demand is imported from the grid, and the excess of each stream is respectively unused or exported. Source: author, schematic.

electricity purchase tariff, surplus electricity at the feed-in tariff, and matched thermal energy as the natural-gas consumption avoided by the existing boiler,

$$S_{\text{industry}} = E_{\text{el,matched}}c_{\text{buy}} + E_{\text{el,export}}c_{\text{export}} + \frac{3.6E_{\text{th,matched}}}{\eta_{\text{boiler}}}c_{\text{gas}}, \quad (34)$$

where the energy terms are annual totals in kWh and the factor 3.6 converts thermal energy from kWh to MJ. Excess thermal energy was assigned no economic value because thermal storage was not modelled. The supply path instead valued the complete annual generation of both streams net of operating cost,

$$B_{\text{net}} = E_{\text{el}}c_{\text{buy}} + \frac{3.6E_{\text{th}}}{\eta_{\text{boiler}}}c_{\text{gas}} - C_{\text{O\&M}}, \quad (35)$$

and supplied the investment indicators below. Because B_{net} assumes complete self-consumption of both streams, the resulting payback, net present value and levelised costs are upper-bound estimates rather than demand-matched results³⁷.

Initial capital expenditure and annual operating expenditure were calculated from

$$C_0 = c_A A, \quad C_{\text{O\&M}} = r_{\text{O\&M}} C_0, \quad (36)$$

where c_A is the installed cost per unit collector area and $r_{\text{O\&M}}$ is the annual operating-cost fraction. Simple payback period and net present value over an analysis period of N years were calculated as^{29,46}

$$\text{SPP} = \frac{C_0}{B_{\text{net}}}, \quad (37)$$

$$\text{NPV} = -C_0 + B_{\text{net}} \left[\frac{1 - (1+i)^{-N}}{i} \right], \quad (38)$$

where i is the discount rate; the implementation substitutes the undiscounted limits when $i = 0$, as set out in Appendix C. Annual savings were assumed to remain constant; energy-price escalation, system degradation, taxation and financing were not modelled.

The calculator also reported levelised costs of electricity and heat using the capital recovery factor

$$\text{CRF} = \frac{i(1+i)^N}{(1+i)^N - 1}. \quad (39)$$

Annualised system costs were allocated between electricity and heat in proportion to their annual generation, one kWh of heat being weighted equally with one kWh of electricity. The complete LCOE and LCOH equations and the cost-allocation method are provided in Appendix C.

Avoided operational emissions were calculated as

$$M_{\text{avoided}} = \frac{E_{\text{el,matched}} EF_{\text{grid}} + \left(\frac{3.6 E_{\text{th,matched}}}{1000 \eta_{\text{boiler}}} \right) EF_{\text{gas}}}{1000}, \quad (40)$$

where EF_{grid} is the selected electricity factor in kg CO₂-e kWh⁻¹, EF_{gas} is the natural-gas factor in kg CO₂-e GJ⁻¹, and M_{avoided} is reported in tonnes CO₂-e per year. The implementation used the Australian National Greenhouse Accounts combined factor of 51.53 kg CO₂-e GJ⁻¹ for pipeline natural gas³¹.

Only directly matched electricity and heat were included in avoided emissions, in contrast to the supply-path financial indicators. Electricity exports were excluded to avoid assuming that all exported generation displaced grid electricity at the selected facility-level emissions factor.

J. Design explorer and system sizing

The Design Explorer was developed to examine how collector area affects thermal-energy utilisation and demand coverage. Collector area A was treated as the principal sizing variable, while the site weather, collector orientation, thermal-model coefficients, inlet-water temperature and facility-demand profile were held constant.

For each candidate collector area, the calculator repeated the complete 8,760-hour PVT thermal calculation using the same thermal function as the principal simulation, rather than proportionally scaling one annual result, and rematched the resulting hourly thermal supply against the stored facility demand using the procedure of Section III.H³⁷.

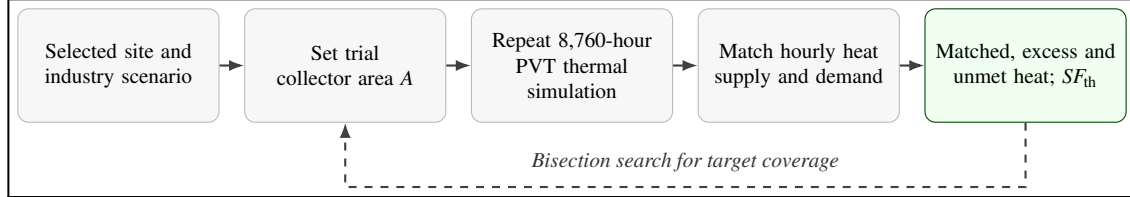


Figure 7: Design Explorer calculation sequence. Collector area is adjusted until the selected thermal solar-fraction target is approached.

The principal sizing outputs were

$$Q_{\text{matched}}(A) = \sum_{h=1}^{8760} \min [Q_{\text{PVT},h}(A), Q_{\text{demand},h}], \quad (41)$$

$$SF_{\text{th}}(A) = \frac{Q_{\text{matched}}(A)}{\sum_{h=1}^{8760} Q_{\text{demand},h}}, \quad (42)$$

Annual excess and unmet thermal energy were also recalculated for each trial area.

The calculator could additionally estimate the collector area required to reach a user-defined target thermal solar fraction SF_{target} , constrained to between 1 and 99%. The upper bound of the search interval was first expanded geometrically until the target was bracketed or a maximum trial area was reached, after which a bisection search performed a fixed fourteen interval halvings, retaining the half in which the target lay. This provided a stable numerical estimate without requiring an analytical sizing equation³⁷.

The Design Explorer provides comparative early-stage sizing guidance rather than a final system design. It does not account for available roof geometry, discrete collector quantities, hydraulic pressure losses, pump selection, pipe sizing, thermal storage or detailed control strategies. The resulting area should therefore be interpreted as an indicative collector requirement for further engineering assessment.

K. Software implementation

The frontend introduced in Section III.A uses HTML, CSS and vanilla JavaScript with no framework or compilation stage, so that it can be deployed directly as static web content^{37,44}.

After the site weather data and mains-water profile have been loaded, the annual calculation routine validates the user inputs and processes the 8,760 hourly records sequentially, evaluating plane-of-array irradiance, PVT thermal and electrical supply and simultaneous facility demand at each timestep. The resulting arrays are stored in a common result object rather than being recalculated separately for each display component, ensuring that all displayed and exported values originate from the same completed annual simulation³⁷.

Source code, model constants, dependencies, documentation and validation material are maintained in a version-controlled GitHub repository³⁷. The application version and repository commit used for the final case studies were recorded so that the reported results can be traced to a defined software state. The repository and deployment structure is summarised in the appendix.

L. Validation and verification

A layered verification and validation procedure was applied to the calculator. Verification assessed whether the implemented software reproduced the intended equations, calculation sequence and data-handling policies. Validation assessed agreement with independent software, reference datasets and available field observations.

At the component level, automated tests checked the solar-geometry equations, irradiance transposition, PVT thermal and electrical calculations, mains-water model, industry-demand functions, hourly energy balances and temporal aggregation. Boundary tests also checked night-time operation, zero irradiance, zero flow, invalid inputs and other conditions that could otherwise produce non-physical or undefined values. Equation-source and numerical locks were used for the frozen PVT models to identify unintended changes to model coefficients or formulation³⁷.

The weather backend was tested separately from the browser calculation. These tests checked the expected record count, required meteorological variables, true-solar-time calculation, local-standard-time demand schedule and frontend-backend data contract. Locked weather fixtures were used during regression testing so that changes in the external PVGIS service did not alter the verification baseline. Browser-level tests subsequently checked that a complete scenario could be loaded, calculated and displayed without calculation or interface failure.

Differences from the independent pvlib, PVGIS and PVWatts comparisons that were attributable

Model or function	Verification method	Validation or comparison evidence
Weather and time handling	Backend contract, field and 8,760-record tests	PVGIS/pvlib source consistency
Solar geometry and POA irradiance	Equation and numerical regression tests	Independent pvlib calculations
PV electrical output	Temperature, loss and aggregation tests	PVGIS and PVWatts reasonableness checks
PVT thermal output	Equation locks and boundary tests	SOAC field-data comparison
BC-Aus mains temperature	Identity, fitting and propagation tests	CER profiles and EnergyPlus ground-temperature proxies
Industry demand and energy matching	Unit, schedule and energy-balance tests	Published benchmarks and process reasonableness checks
Complete web application	Browser workflow and no-error tests	Reproduction of locked case-study scenarios

Table 6: Verification and validation evidence applied to the principal calculator components.

to the calculator’s isotropic diffuse-sky model were treated as a known modelling choice rather than a software error.

The BC-Aus mains-water model was assessed against the legacy Clean Energy Regulator monthly reference profiles used during parameter fitting. An independent physical cross-check was also performed against EnergyPlus undisturbed ground-temperature profiles at 0.5 and 2.0 m depth. These ground-temperature values were treated only as proxies for buried mains-water behaviour because they do not represent measured water temperatures. Automated tests additionally checked reference identity, deterministic coefficient regeneration, runtime reference selection and propagation of mains temperature into the PVT inlet and industry water-heating calculations.

Available field validation used 19 days of five-minute operating data from the Sydney Olympic Aquatic Centre installation. Measured thermal behaviour was compared with the collector model under corresponding operating conditions. The field comparison was used to evaluate the practical representativeness of the coefficient-based thermal model rather than to refit its coefficients.

The numerical results of these checks are presented in Sections IV.A and IV.B. The evidence summarised in Table 6 does not establish universal field accuracy: no measured Australian mains-water dataset was available, the SOAC dataset covered a limited operating period, and the simplified electrical cooling adjustment was not independently validated. The complete test inventory, fixture descriptions and detailed pass/fail outputs are provided in the appendix.

IV. Results and discussion

A. Software and model verification

Effect of the cooling assumption. Figure 8 isolates the Section E. cooling assumption on the clearest day of the Sydney typical meteorological year. Circulating water held the modelled cell temperature 28.9 K below the uncooled case at the midday peak, 31.2°C against 59.4°C. Because the cell-temperature correction reduces electrical efficiency as the cell warms, that temperature difference raised alternating-current output by 13.5% in the peak hour and by 9.6% over the day. Across the full year the same effect produced 6042 kWh against 5657 kWh, a gain of 6.8%, because most hours are cooler and less sunny than the day shown.

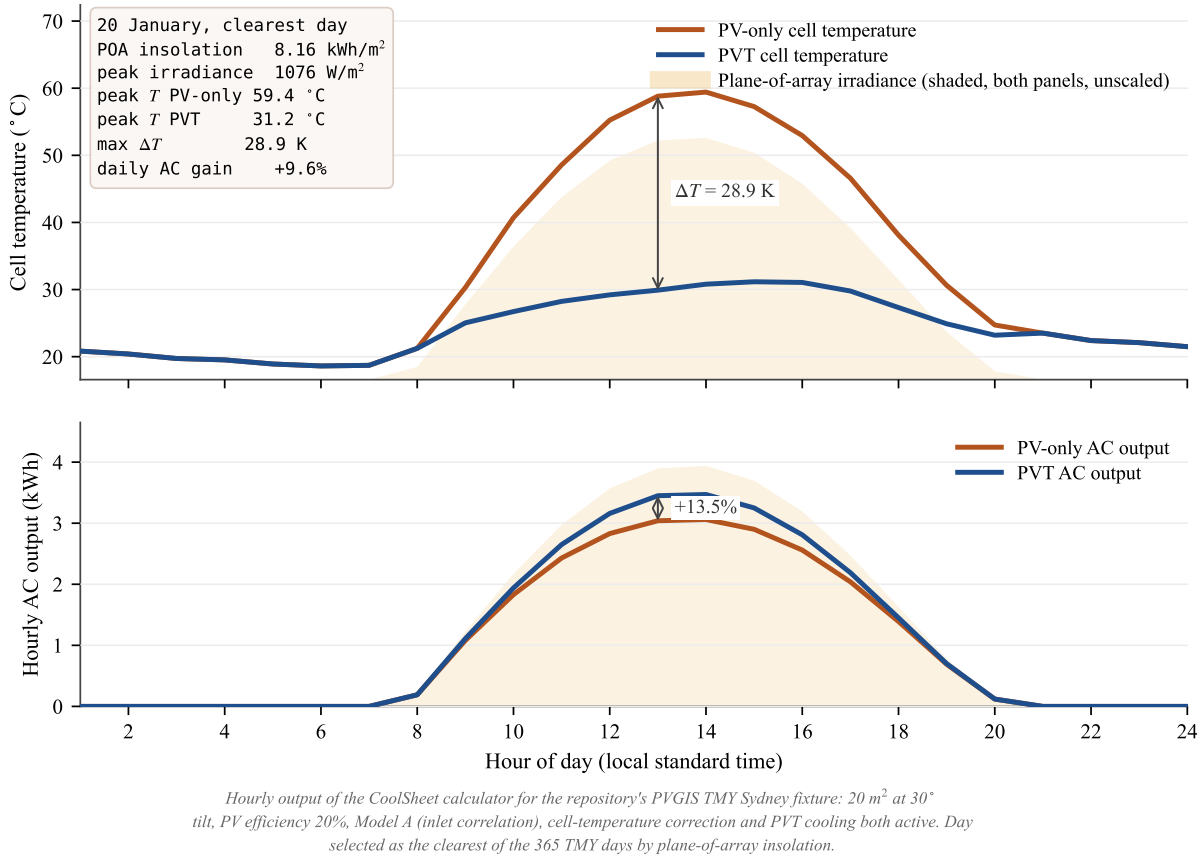


Figure 8: Modelled cell temperature and alternating-current output with and without PVT cooling, for the clearest day of the Sydney typical meteorological year. Both panels share the hourly time axis and are taken from a single run of the calculator.

B. Mains-water temperature results

Development of the final parameterisation. Retaining the original +6°F offset within a single national Australian fit produced an overall RMSE of 4.590°C against the five CER reference

profiles. Removing the fixed offset reduced the RMSE to 3.556°C, confirming that the original American offset was not generally suitable across the Australian reference climates.

However, removing the offset only shifted the average level of the predicted profile. It did not account for differences between locations in seasonal amplitude and timing. The final reference-specific fitting approach reduced the overall in-sample RMSE to 0.705°C by allowing the offset, amplitude ratio and phase lag to vary between references. Table 7 summarises this progression.

Model formulation	Overall RMSE (°C)
National fit with fixed +6°F offset	4.590
National fit without fixed offset	3.556
Reference-specific BC-Aus fit	0.705

Table 7: Development of the BC-Aus parameterisation against the CER reference profiles.

Agreement with the CER reference profiles. The final BC-Aus formulation reproduced the monthly reference profiles with RMSE values ranging from 0.502°C for Melbourne to 0.950°C for Alice Springs. The maximum monthly absolute deviation among the four runtime references was 1.636°C. Table 8 lists the location-specific error statistics and Figure 9 shows the corresponding monthly comparison.

Reference location	MAE (°C)	RMSE (°C)	Maximum absolute error (°C)
Rockhampton	0.667	0.697	0.998
Alice Springs	0.779	0.950	1.636
Sydney	0.523	0.602	1.251
Melbourne	0.496	0.502	0.613

Table 8: Agreement between BC-Aus and the CER monthly reference profiles.

These results demonstrate that the fitted sinusoidal model closely reproduces the available reference profiles. They do not demonstrate equivalent accuracy against actual water temperatures because the CER values were used during model calibration.

Independent physical cross-check. The final profiles were compared with EnergyPlus undisturbed ground-temperature estimates at 0.5 m and 2.0 m depth. The 0.5 m profiles generally exhibited greater seasonal variation, whereas the deeper profiles showed greater damping. Comparison with both depths provided a useful independent check of seasonal timing and temperature range because the EnergyPlus data were not used during coefficient fitting. EnergyPlus documentation

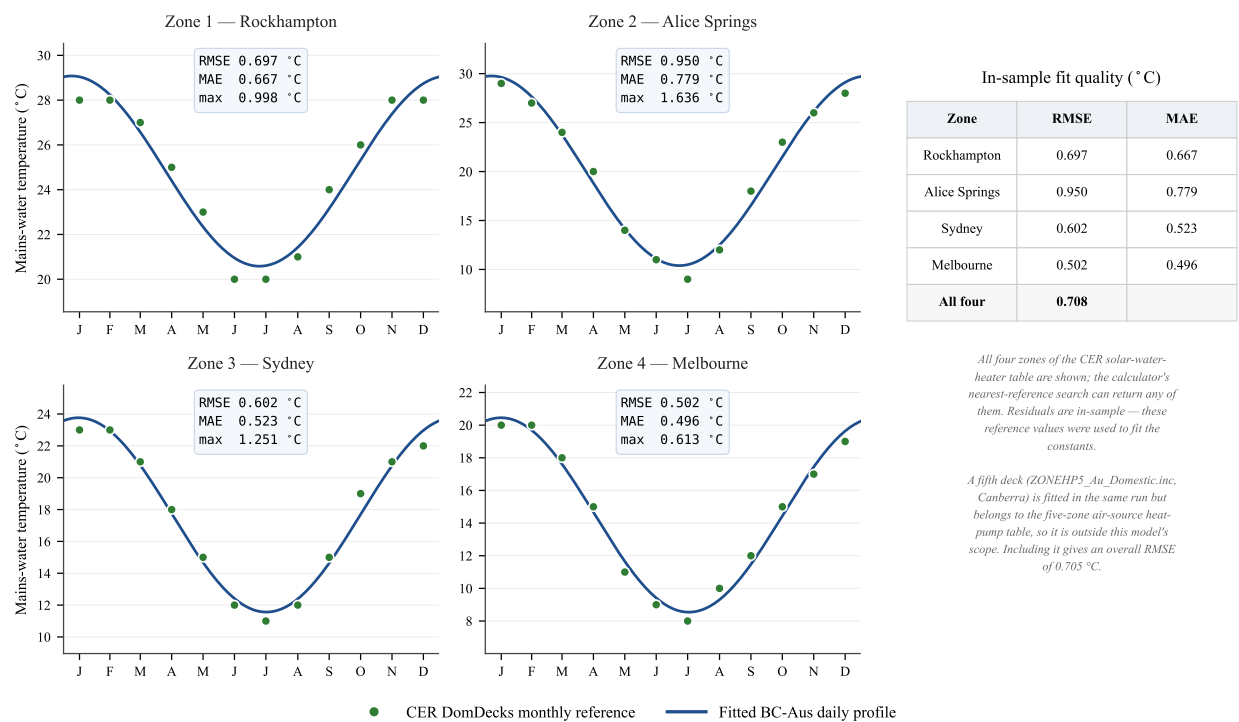


Figure 9: Fitted BC-Aus daily mains-water profiles against the twelve CER monthly reference values for each of the four solar-water-heater reference locations. Agreement is in-sample: the plotted reference values were used to fit the coefficients.

identifies the shallow-ground values as calculated undisturbed temperatures at approximately 0.5 m depth rather than measured mains-water data.

[Figure IV.Y to be inserted here – optional. Comparison of BC-Aus mains-water temperature predictions with EnergyPlus undisturbed ground-temperature estimates at 0.5 m depth, presented as an independent physical proxy rather than measured mains-water temperature. Include only if it fits comfortably within the page limit; requires the actual EnergyPlus comparison data, which has not been supplied yet. The full 2.0 m comparison can remain in the repository validation material or an appendix instead of the main report.]

Software verification. Automated testing confirmed that the correct source identities were associated with the fitted references, the production coefficients could be regenerated deterministically and only the four intended solar-water-heater references could be selected at runtime. Downstream tests also confirmed that colder daily mains-water temperatures increased water-heating demand while warmer temperatures reduced it, without affecting unrelated demand components.

Limitations and interpretation. No measured Australian mains-water dataset was available for independent validation. The overall RMSE of 0.705°C therefore represents agreement with the legacy CER reference profiles used during fitting and must not be interpreted as field prediction accuracy.

The runtime model also uses the geographically nearest of four reference locations, which may not always be the most climatologically similar reference. Furthermore, the fitted coefficients were derived using the historical weather basis of the CER decks, while runtime calculations use PVGIS TMY statistics for the selected site. EnergyPlus ground temperatures provide an independent physical comparison but remain calculated soil-temperature proxies and do not account for water source, pipe materials, burial geometry, residence time or network operation.

The BC-Aus model closely reproduces the available Australian CER cold-water reference profiles and exhibits physically plausible seasonal behaviour relative to independent ground-temperature proxies. Validation against measured mains-water temperatures at independent Australian sites remains necessary before field prediction accuracy can be established.

C. Industry-demand model results

D. Principal case study 1

E. Principal case study 2

F. Cross-industry comparison

G. Sensitivity and sizing analysis

H. Limitations and uncertainty

I. Engineering significance

V. Conclusions

A. Main findings

B. Contributions

C. Practical value

D. Recommendations and future work

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NOTE – entries 1–46 were checked against the published record on 27 July 2026: author lists, titles, volume/article numbers, DOIs and URLs verified. Still to confirm personally: [6] Fu’s exact thesis title, and [37] the CoolSheet version and commit identifier once the final case studies are fixed. Delete this note before submission.

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Appendices

Appendix A — Calculator input register, model coefficients and solver configuration

Tables 9–12 record every parameter the calculator reads, with its interface label, the identifier used in the source, units, the value shipped with the application and the accepted range or validation rule. *User* denotes a value entered directly, *derived* a value the software computes from other inputs, and *optional* a toggle or override that is inactive unless selected. Ranges given as “clamped” are silently constrained to the stated interval rather than rejected; ranges given as limits are rejected with a message. Blank or non-numeric entries fall back to the shipped default, except where a fallback is stated separately.

Interface label	Identifier	Unit	Default	Range or rule	Type
Address	addressInput	–	(blank)	geocoded; warning if outside Australia	user
Collector / PV area	area	m ²	250	> 0	user
Panel tilt	tiltAngle	°	30	0–90	user
Panel direction	azimuthAngle	°	0	–180 to +180; 0° = north	user
Ground reflectivity	albedo	–	0.2	0–1	user
Water flow rate	flowRate	L s ^{–1} m ^{–2}	0.02	> 0	user
Panel efficiency	etaPvPercent	%	20	0–100	user
PV temperature loss	pvtTempCoeff	% °C ^{–1}	–0.40	finite; –0.40 if blank	user
NOCT panel temperature	pvtNoct	°C	45	20–80	user
Non-inverter system losses	pvtSystemLossPct	% of DC	14	clamped 0–99	user
Inverter efficiency	pvtInverterEfficiencyPct	%	96	clamped 1–100	user
Module temperature correction	pvtTempCorrEnable	–	on	forced on outside diagnostic mode	optional
PVT cooling sensitivity	pvtCoolingSensitivityEnable	–	on	forced on outside diagnostic mode	optional
Thermal model	thermalModel	–	A	A or B	user
Custom monthly mains temperatures	mainsCustomEnable	–	off	on or off	optional
Monthly mains temperature	mainsM0–mainsM11	°C	from model	blank months use the modelled daily value	optional
Set all months to	mainsQuickSet	°C	(blank)	finite; enables custom mode	optional
Diagnostic mode	chkHideMains	–	off	on or off	optional

Table 9: Site, PVT system and mains-override input register.

Table 10: Industry-demand input register. Process-selection checkboxes are additionally provided for each industry, all selected by default.

Interface label	Identifier	Unit	Default	Range or rule	Type
<i>All industries</i>					
Industry	industrySelect	–	none	one of five models	user
Operating profile	profileType	–	24/7	continuous or Mon–Fri	user
Throughput per year	throughputInput	L yr ^{−1}	5,000,000	> 0; dairy and brewery only	user
<i>Dairy farm</i>					
Electricity intensity	dairyElectricKWhPerKL	kWh kL ^{−1}	51.7	≥ 0	user
Fatty-film rinse water	dairyFattyWater	LL ^{−1}	0.30	≥ 0	user
CIP preheat water	dairyCipWater	LL ^{−1}	0.57	≥ 0	user
Boiler preheat water	dairyBoilerWater	LL ^{−1}	0.50	≥ 0	user
Preheat target	dairyTargetTemp	°C	35	clamped 15–95; shared by all three processes	user
<i>Brewery</i>					
Electricity intensity	breweryElectricKWhPerHL	kWh hL ^{−1}	11.5	≥ 0	user
CIP pre-rinse water	breweryCipWater	LL ^{−1}	0.80	≥ 0	user
Bottle/keg rinse water	breweryRinseWater	LL ^{−1}	0.45	≥ 0	user
Boiler preheat water	breweryBoilerWater	LL ^{−1}	0.60	≥ 0	user
CIP/boiler target	breweryCipTarget	°C	45	clamped 15–95	user
Bottle/keg rinse target	breweryRinseTarget	°C	40	clamped 15–95	user
<i>Hotel</i>					
Total available rooms	hotelRoomsInput	–	147	> 0	user
Occupancy rate	hotelOccupancyInput	%	73.4	0–100	user
Occupied room-nights	(derived)	yr ^{−1}	39,383	rooms × 365 × occupancy	derived
Domestic hot water	hotelDhwKWh	kWh unit ^{−1}	4.5	≥ 0	user
Kitchen/dishwashing	hotelKitchenKWh	kWh unit ^{−1}	1.6	≥ 0	user
Laundry	hotelLaundryKWh	kWh unit ^{−1}	1.2	≥ 0	user
Pool heating	hotelPoolKWh	kWh unit ^{−1}	4.2	≥ 0	user
Whole-site electricity	hotelElectricKWh	kWh unit ^{−1}	15	≥ 0	user
Metered annual fuel	hotelMeasuredAnnualFuelGj	GJ	(blank)	≥ 0	optional
Apply meter calibration	hotelApplyMeterCalibration	–	off	on or off	optional
Monthly metered fuel	hotelMeterJanFuelGj...	GJ	(blank)	twelve fields, ≥ 0	optional
<i>Aquatic centre</i>					
Indoor pool area	aquaticIndoorArea	m ²	350	≥ 0	user
Outdoor pool area	aquaticOutdoorArea	m ²	250	≥ 0	user
Kids pool area	aquaticKidsArea	m ²	90	≥ 0	user
Sauna area	aquaticSaunaArea	m ²	25	≥ 0	user
Pool cover off-hours	aquaticPoolCover	–	on	on or off	optional
Electricity assumption	aquaticElectricKWhPerM2	kWh m ^{−2} yr ^{−1}	250	≥ 0	user

Interface label	Identifier	Unit	Default	Range or rule	Type
Evaporation multiplier	aquaticEvaporationScale	–	1	clamped 0–3	user
Makeup-water multiplier	aquaticMakeupScale	–	1	clamped 0–3	user
<i>Commercial laundry</i>					
Laundry processed	laundryKgPerDay	kg day ^{−1}	1,500	≥ 0	user
Operating days per week	laundryOperatingDaysPerWeek	–	6	clamped 0–7	user
Wash hot-water target	laundryWashTempC	°C	60	clamped 20–95	user
Total wash water	laundryWaterUseLPerKg	L kg ^{−1}	10	≥ 0	user
Water-use scenario	laundryWaterScenario	L kg ^{−1}	custom	presets 10, 12, 15, 17, 22	optional
Hot-water fraction	laundryHotWaterFraction	–	0.65	clamped 0–1	user
Warm-rinse fraction	laundryWarmRinseFraction	–	0.20	clamped 0–1	user
Warm-rinse target	laundryWarmRinseTempC	°C	35	clamped 15–60	user
System loss fraction	laundrySystemLossFraction	–	0	clamped 0–1	user

Interface label	Identifier	Unit	Default	Range or rule	Type
Electricity bought from grid	electricityPrice	AUD kWh ⁻¹	0.27	≥ 0	user
Electricity exported to grid	feedInTariffInput	AUD kWh ⁻¹	0.06	≥ 0	user
Natural gas price	gasPriceInput	AUD MJ ⁻¹	0.020	≥ 0	user
Existing boiler efficiency	boilerEffInput	–	0.85	clamped 0.5–1	user
Region for emissions estimate	gridEmissionFactor	kg CO ₂ -e per kWh	0.62	0.20–0.78 by region	user
Fill installed cost from \$/W	autoCapexFromWatts	–	on	on or off	optional
PV installed cost	pvInstalledCostPerW	AUD W ⁻¹	1.20	≥ 0	user
Thermal / PVT installed cost	thermalInstalledCostPerW	AUD W ⁻¹	1.50	≥ 0	user
Installed cost	capexInput	AUD m ⁻²	800	≥ 0 ; overwritten with 540 while auto-fill is on	user or derived
Calculated CAPEX	calculatedCapexPerM2	AUD m ⁻²	540	read-only	derived
Installed cost conversion	installedCostPerFt2	AUD ft ⁻²	50.17	read-only	derived
Annual operating cost	opexRateInput	% of C_0	1.5	0–20	user
System lifetime	systemLifeInput	yr	25	1–50, floored to integer	user
Discount rate	discountRateInput	%	6	0–30	user
Target heat coverage	designExplorerTargetInput	%	from result	clamped 1–99	user

Table 11: Economics, emissions, finance and sizing input register.

Quantity	Symbol	Stored value	Unit
<i>Model A — inlet-temperature correlation</i>			
Zero-loss coefficient	$a_{0,A}$	0.279952866	–
Inlet-temperature loss coefficient	$a_{1,A}$	–10.52839866	$\text{W m}^{-2}\text{K}^{-1}$
Wind-speed coefficient	$a_{2,A}$	–0.008135537	s m^{-1}
<i>Model B — ISO 9806 formulation</i>			
Zero-loss efficiency	η_0	0.762	–
Linear heat-loss coefficient	a_1	3.93	$\text{W m}^{-2}\text{K}^{-1}$
Quadratic heat-loss coefficient	a_2	0.0095	$\text{W m}^{-2}\text{K}^{-2}$
Wind×temperature loss	a_3	0	$\text{J m}^{-3}\text{K}^{-1}$
Sky-temperature coefficient	a_4	0	–
Wind×irradiance coefficient	a_6	0	s m^{-1}
Fourth-order loss coefficient	a_8	0	$\text{W m}^{-2}\text{K}^{-4}$
<i>Newton–Raphson solver (Model B only)</i>			
Initial outlet estimate	$T_{\text{out}}^{(0)}$	40	$^{\circ}\text{C}$
Iteration cap	–	5	interface range 1–10
Convergence tolerance on ΔT_{out}	–	1×10^{-4}	$^{\circ}\text{C}$
Minimum Jacobian magnitude	–	1×10^{-12}	W K^{-1}
<i>Fixed physical constants</i>			
Water specific heat	c_p	4184	$\text{J kg}^{-1}\text{K}^{-1}$
Water specific heat (demand and outlet terms)	c_p	4.184	$\text{kJ kg}^{-1}\text{K}^{-1}$
Water density (implicit, not a named constant)	ρ_w	1	kg L^{-1}
Stefan–Boltzmann constant	σ	5.67×10^{-8}	$\text{W m}^{-2}\text{K}^{-4}$
Sky long-wave irradiance	E_L	$5.31 \times 10^{-13} T_{a,K}^6$	W m^{-2}
Standard test cell temperature	–	25	$^{\circ}\text{C}$
NOCT reference irradiance and ambient	–	800 and 20	W m^{-2} and $^{\circ}\text{C}$
Latent heat of evaporation (aquatic)	–	0.680	kWh kg^{-1}
Natural-gas emissions factor	EF_{gas}	51.53	$\text{kg CO}_2\text{-e GJ}^{-1}$
Energy unit conversion	–	1000/3.6	kWh GJ^{-1}
Area unit conversion	–	10.76391041671	$\text{ft}^2 \text{m}^{-2}$
<i>Numerical thresholds</i>			
Minimum irradiance for thermal output	–	1×10^{-6}	W m^{-2}
Minimum mass flow for thermal output	–	1×10^{-12}	kg h^{-1}
Minimum U_L for the cooling term	–	1×10^{-6}	$\text{W m}^{-2}\text{K}^{-1}$
Daytime temperature sample window	–	$G > 50$, hours 10–16	W m^{-2} , local standard time

Table 12: Thermal-model coefficients, solver configuration, fixed physical constants and numerical thresholds, at full stored precision. All Model A and Model B coefficients are editable in the interface; the constants and thresholds are not.

The Model A and Model B coefficient sets are not interchangeable: Model A is evaluated on inlet-fluid temperature and Model B on mean-fluid temperature. Setting a_3 , a_4 , a_6 and a_8 to zero reduces the implemented ISO 9806 expression to the quadratic form given as Eq. 13; the terms remain in the software so that a complete certified coefficient set can be substituted. The weather backend advertises the long-wave term as prohibited for the frozen production model.

Code and repository information

The source code for the CoolSheet PVT Calculator is publicly available at:

GitHub repository:

<https://github.com/coolsheet-pvt/coolsheet-pvt.github.io>

Live calculator:

<https://coolsheet-pvt.github.io/>

The repository contains the frontend application, calculation code, weather backend, validation files and supporting documentation. The public frontend is hosted using GitHub Pages⁴⁵, which publishes the repository’s static HTML, CSS and JavaScript files, while the Python FastAPI weather service is hosted separately on Render. Table 13 summarises this structure.

Because the software may continue to be updated after submission, the repository should be accessed alongside the thesis methodology and stated model assumptions when reproducing the reported results.

BC-Aus calculation workflow

Figure 10 summarises the complete BC-Aus calculation workflow described in Section III.F.

Weather-data record structure (Appendix B)

Table 14 lists the fields retained in each hourly weather record after validation and normalisation, as described in Section III.C.

The frontend refuses a weather response that does not declare data-contract version 2.1 and PVGIS API version 5.3, that does not contain exactly 8,760 records, that omits or falsifies any field in Table 14, or whose day-hour and UTC keys are not each unique across all 8,760 hours. Retrieval uses the endpoint https://re.jrc.ec.europa.eu/api/v5_3/ with `startyear=2005`, `endyear=2023`, us

Component	Principal files or service	Function
User interface	index.html, CSS files	Inputs, controls and output containers
Calculation engine	js/app.js	Hourly engineering and financial calculations
Model data	JavaScript constants and validation fixtures	Coefficients, defaults and reference data
Weather backend	pvt-tmy-api/server.py	PVGIS retrieval and temporal preprocessing
Frontend hosting	GitHub Pages	Static application deployment
Backend hosting	Render	FastAPI service deployment
Validation	validation/ directory	Unit, backend and browser tests

Table 13: Repository and deployment structure of the CoolSheet PVT Calculator.

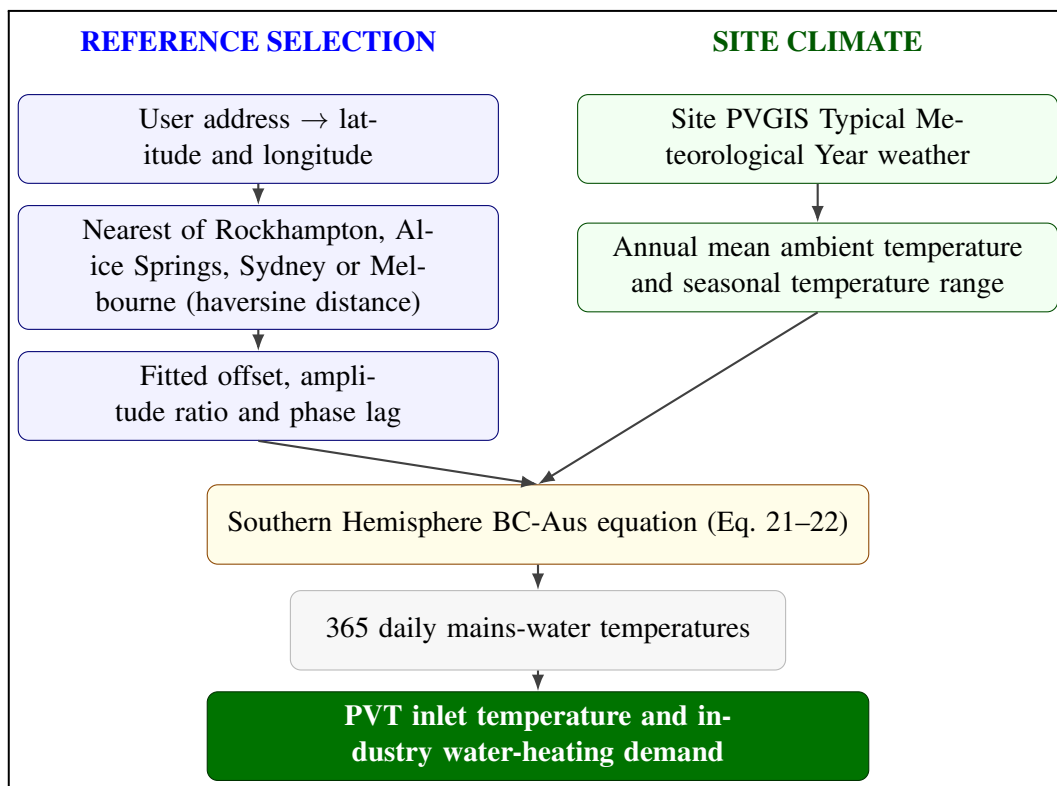


Figure 10: Calculation workflow of the BC-Aus mains-water temperature model. The user's coordinates select the nearest Australian reference coefficients, while site-specific PVGIS weather supplies the ambient-temperature statistics used to generate a 365-day mains-water profile.

Field	Unit	Function
dayN	–	Day-of-year indexing and aggregation
hourN	h	Local-standard-time demand scheduling
solarHour	h	True solar time for solar geometry
utcTimestamp	–	Temporal provenance
dni	W m ⁻²	Direct irradiance input
dhi	W m ⁻²	Diffuse irradiance input
ghi	W m ⁻²	Horizontal irradiance and ground reflection
ta	°C	Ambient-temperature input
vwind	m s ⁻¹	Wind-speed input
relativeHumidityPct	%	Supporting weather field
infraredHorizontalWm2	W m ⁻²	Provenance and supporting analysis

Table 14: Weather-data record structure retained by the CoolSheet PVT Calculator after validation.

ehorizon=true, coerce_year=1990, map_variables=true and JSON output. Each response also carries a provenance block recording the PVGIS database identifier, the months selected for the typical year, the installed versions of pvlib, pandas, NumPy, timezonefinder and tzdata, the record count, the standard UTC offset applied to the demand clock, and a SHA-256 digest of the record array canonicalised as UTF-8 JSON with sorted keys, comma and colon separators and no non-finite values. Responses are cached for 24 hours on the server and seven days in the browser.

Appendix C — Detailed economic and emissions calculations

Annualised cost and allocation. The initial capital cost and annual operating cost follow Eq. 36. The annualised system cost is

$$A_{\text{total}} = C_0 \text{CRF} + C_{\text{O\&M}}, \quad (43)$$

with CRF from Eq. 39. Annual generation is converted to a single energy-equivalent basis and split between the two streams,

$$E_{\text{eq}} = E_{\text{el}} + f_{\text{th} \rightarrow \text{e}} E_{\text{th}}, \quad f_{\text{th} \rightarrow \text{e}} = 1, \quad (44)$$

$$\varphi_{\text{el}} = \frac{E_{\text{el}}}{E_{\text{eq}}}, \quad \varphi_{\text{th}} = 1 - \varphi_{\text{el}}, \quad (45)$$

where E_{el} is annual net alternating-current generation and E_{th} annual useful thermal output, both *gross* rather than demand-matched. The weighting $f_{th \rightarrow e} = 1$ treats one kilowatt-hour of heat as equal in value to one kilowatt-hour of electricity; it is a value assumption and not an exergy or energy-quality weighting. Where E_{eq} is numerically zero the split defaults to $\varphi_{el} = \varphi_{th} = 0.5$.

Levelised costs. The three reported levelised costs are

$$LCOE = \frac{\varphi_{el} A_{total}}{E_{el}}, \quad (46)$$

$$LCOH = \frac{\varphi_{th} A_{total}}{E_{th}}, \quad (47)$$

$$LCOE_{comb} = \frac{A_{total}}{E_{eq}}. \quad (48)$$

Substituting Eq. 45 shows that all three reduce to A_{total}/E_{eq} while $f_{th \rightarrow e} = 1$. The separately reported electrical and thermal levelised costs are therefore numerically identical in the shipped configuration, and differ only if the heat-to-electricity weighting is changed. Each is suppressed rather than reported when the corresponding annual energy is not greater than 1×10^{-9} kWh.

Zero discount rate. A discount rate of zero makes both Eq. 39 and the annuity factor in Eq. 38 indeterminate. For $i \leq 1 \times 10^{-9}$ the implementation substitutes the undiscounted limits

$$CRF = \frac{1}{N}, \quad NPV = -C_0 + B_{net}N, \quad (49)$$

which are the limits of Eqs. 38 and 39 as $i \rightarrow 0$. A rate entered as 0% therefore takes this branch exactly. Simple payback does not depend on i and is reported only where B_{net} exceeds 1×10^{-9} AUD yr⁻¹; a non-positive net benefit is reported as no payback rather than as a negative period.

Worked example. Table 15 follows the shipped dairy scenario from milk throughput to levelised cost. Facility demand and all financial quantities are exact arithmetic on the stated inputs. The two annual supply totals are plausible round values assumed for the illustration, because gross generation depends on the site weather file; they are not simulation output. Tariffs, cost rates and emissions factors are the shipped defaults of Table 11.

Quantity	Value	Basis
<i>Facility demand</i>		
Annual milk throughput	5,000,000 L	shipped default
Total process-water coefficient	1.37 L L ⁻¹	0.30 + 0.57 + 0.50
Annual heated process water	6,850,000 L	5,000,000 × 1.37
Annual thermal demand at 18 to 35 °C	135,300 kWh	Eq. 24
Annual electrical demand	258,500 kWh	51.7 × 5,000
<i>Annual supply (assumed for the illustration)</i>		
Collector area A	250 m ²	shipped default
Net alternating-current generation E_{el}	65,000 kWh	assumed
Useful thermal output E_{th}	105,000 kWh	assumed
Directly matched electricity	41,000 kWh	assumed
Exported electricity	24,000 kWh	65,000 – 41,000
Directly matched heat	47,000 kWh	assumed
<i>Costs</i>		
Installed cost rate c_A	540 AUD m ⁻²	(1.20 + 1.50) × 0.20 × 1000
Capital cost C_0	135,000 AUD	540 × 250
Annual operating cost $C_{O\&M}$	2,025 AUD yr ⁻¹	0.015 × 135,000
Capital recovery factor	0.078227	Eq. 39, $i = 6\%$, $N = 25$
Annualised cost A_{total}	12,586 AUD yr ⁻¹	Eq. 43
<i>Industry path (matched energy)</i>		
Matched electricity value	11,070 AUD yr ⁻¹	41,000 × 0.27
Export value	1,440 AUD yr ⁻¹	24,000 × 0.06
Displaced gas value	3,981 AUD yr ⁻¹	(3.6 × 47,000/0.85) × 0.020
Total industry savings $S_{industry}$	16,491 AUD yr ⁻¹	Eq. 34
Avoided emissions $M_{avoided}$	35.7 t CO ₂ -e yr ⁻¹	Eq. 40
<i>Supply path (gross energy)</i>		
Electricity value at purchase tariff	17,550 AUD yr ⁻¹	65,000 × 0.27
Displaced gas value	8,894 AUD yr ⁻¹	(3.6 × 105,000/0.85) × 0.020
Net annual benefit B_{net}	24,419 AUD yr ⁻¹	Eq. 35
Simple payback period	5.5 yr	Eq. 37
Net present value	177,200 AUD	Eq. 38
Energy-equivalent total E_{eq}	170,000 kWh	Eq. 44
Electrical allocation ϕ_{el}	0.3824	Eq. 45
LCOE, LCOH and combined LCOE	0.0740 AUD kWh ⁻¹	Eqs. 46–48

Table 15: Worked economic example for the shipped dairy scenario, showing the two evaluation paths of Section III.I side by side. The industry path counts only demand-matched energy but does not deduct operating cost; the supply path deducts operating cost but assumes complete utilisation of both streams. The two are therefore not directly comparable, and the supply-path indicators are upper bounds.